

The background of the slide is a dark blue field filled with a complex network of thin, glowing green lines. These lines are interconnected and radiate from various points, creating a sense of dynamic energy and connectivity. Scattered throughout this network are numerous small, bright yellow and orange circular dots, some of which appear to be at the ends of the green lines, resembling nodes in a network or particles in motion.

Energia e Sustentabilidade: duas faces da mesma moeda?

Rui M. Novais

Energia e sustentabilidade



<https://www.youtube.com/watch?v=LwjC073Y9oc>

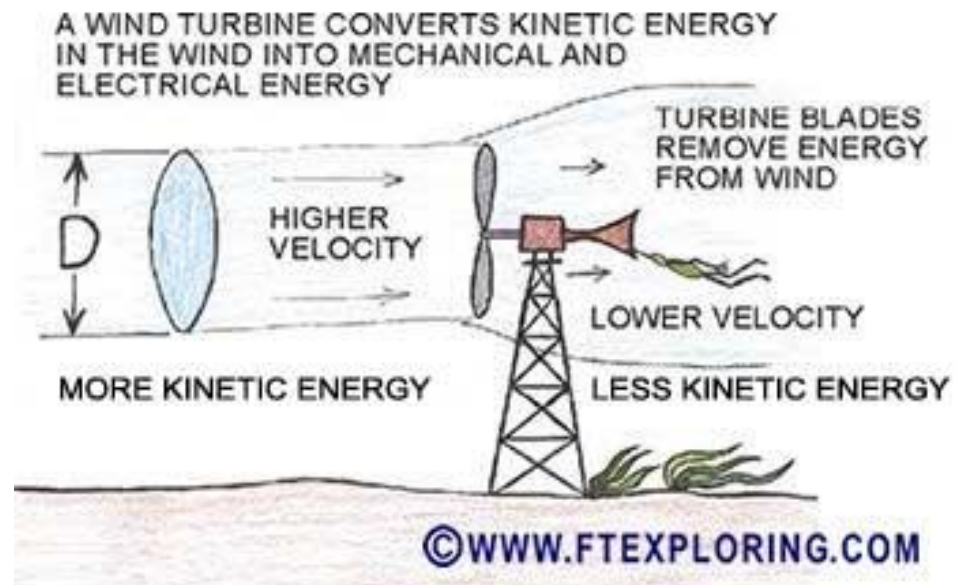
Energia e sustentabilidade

Conceitos fundamentais

Lei da conservação da energia – a quantidade total de energia no Universo permanece constante.

- **1ª lei da termodinâmica:** a energia pode ser convertida de uma forma para outra, mas não pode ser criada ou destruída.

$$\Delta E = q + w$$



Energia e sustentabilidade

Conceitos fundamentais

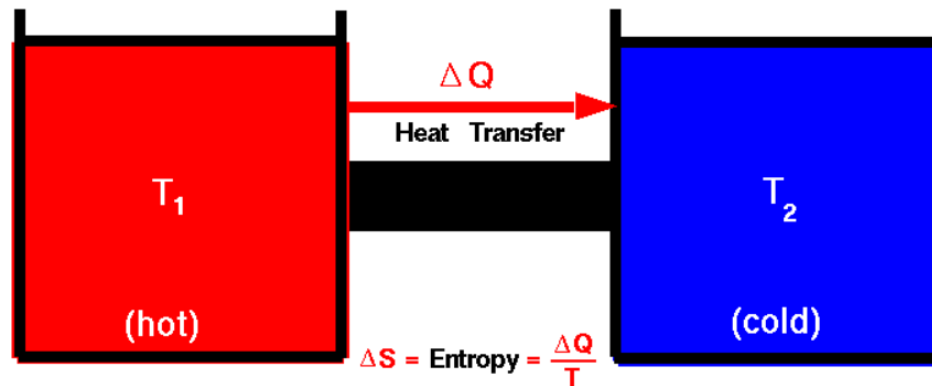
- **2ª lei da termodinâmica:** a entropia do Universo aumenta numa transformação espontânea e mantém-se constante numa situação de equilíbrio.

Num processo espontâneo:

$$\Delta S > 0$$

Situação de equilíbrio:

$$\Delta S = 0$$



There exists a useful thermodynamic variable called entropy (S). A natural process that starts in one equilibrium state and ends in another will go in the direction that causes the entropy of the system plus the environment to increase for an irreversible process and to remain constant for a reversible process.

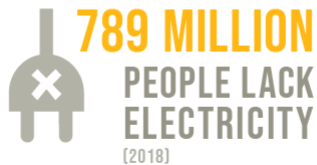
$$S_f = S_i \text{ (reversible)}$$

$$S_f > S_i \text{ (irreversible)}$$

Energia e sustentabilidade

O progresso das sociedades está intrinsecamente associado à energia. O acesso a energia limpa e de baixo custo é um assunto crítico para a criação de melhores condições de vida!

EFFORTS NEED **SCALING UP**
ON SUSTAINABLE ENERGY

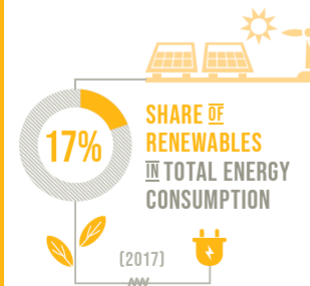


COVID-19 IMPLICATIONS

AFFORDABLE AND RELIABLE ENERGY
IS CRITICAL FOR HEALTH FACILITIES



STEPPED-UP EFFORTS
IN RENEWABLE ENERGY
ARE NEEDED



ENERGY EFFICIENCY
IMPROVEMENT RATE
FALLS SHORT OF
3% NEEDED



789 million
people around
the world lack
access to
electricity

FINANCIAL FLOWS TO DEVELOPING COUNTRIES
FOR RENEWABLE ENERGY ARE INCREASING

\$21.4
BILLION
(2017)



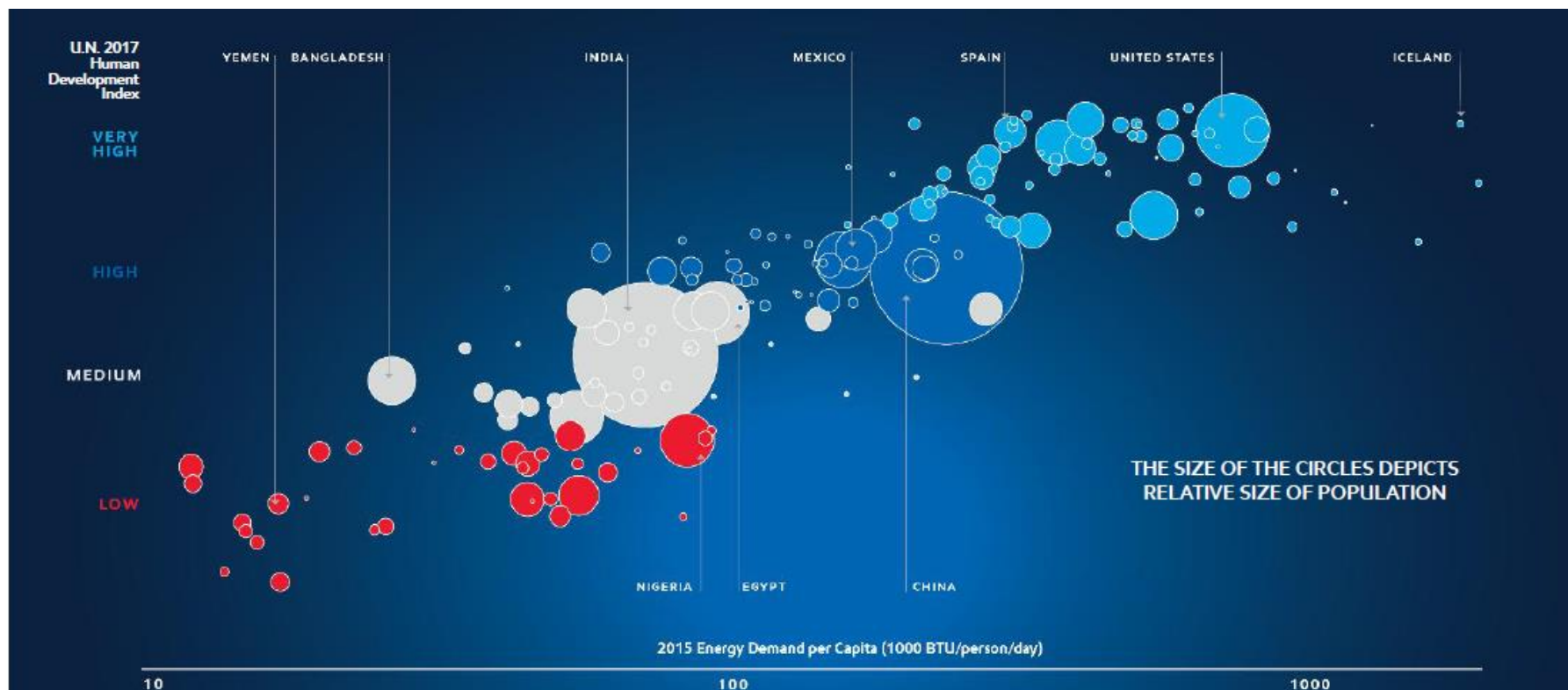
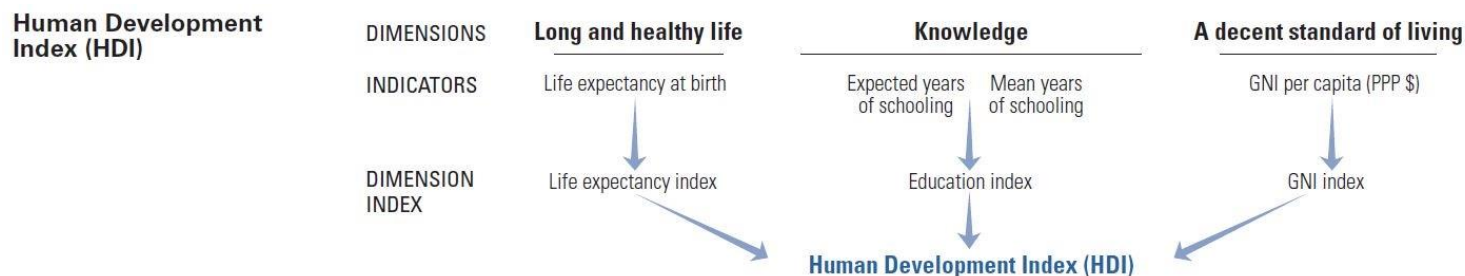
BUT ONLY 12% GOES TO
LDCs

Fonte:

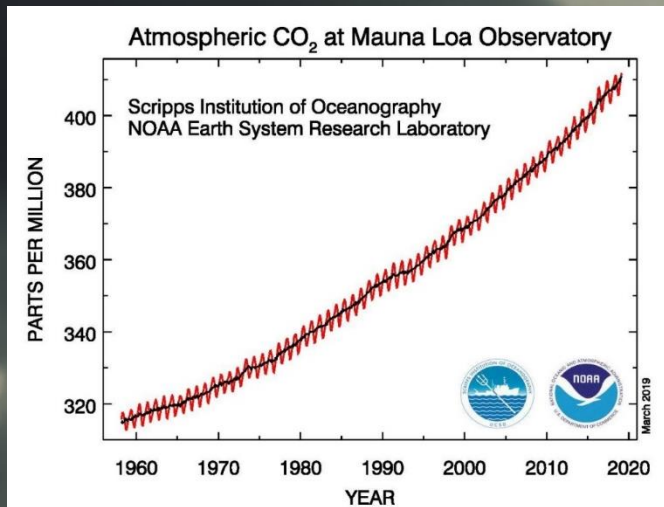
<https://www.un.org/sustainabledevelopment/energy/>

Energia e sustentabilidade

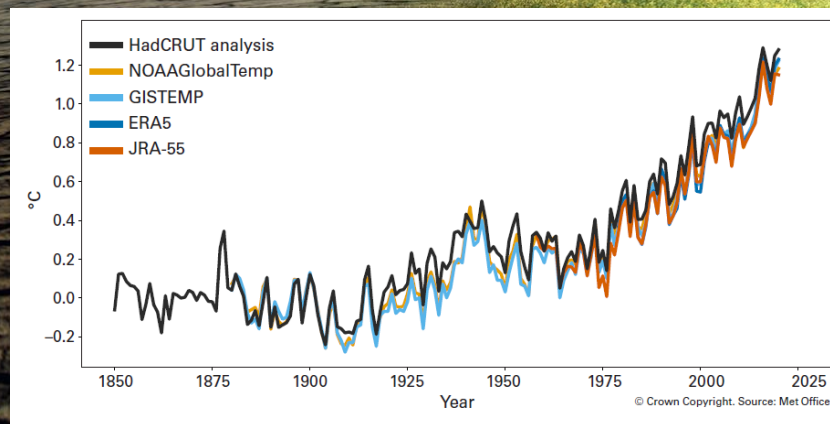
Human development Index (ONU)



Where we are...

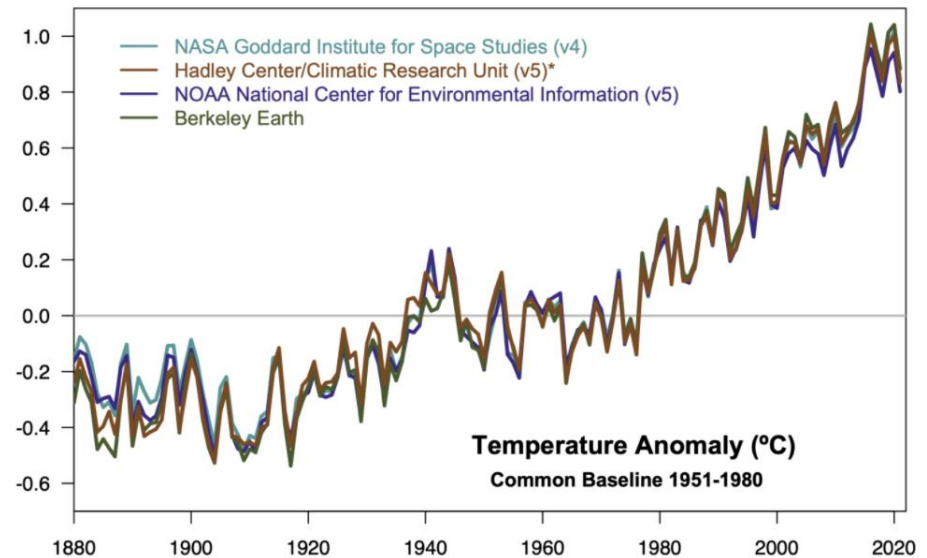
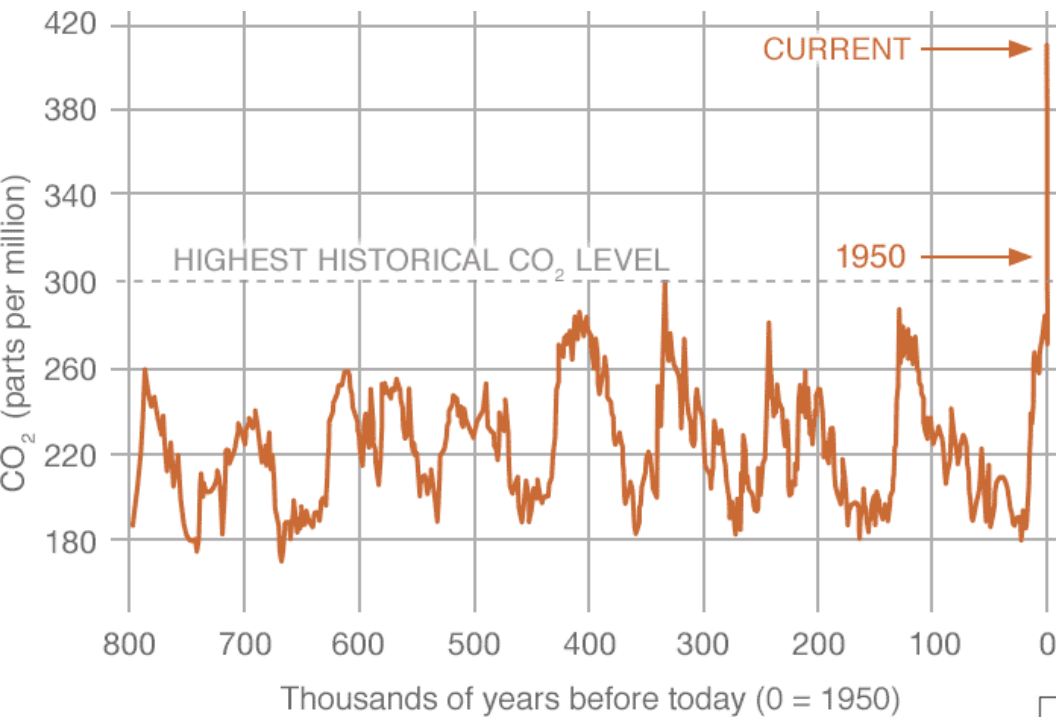


Retirado de: https://climate.nasa.gov/internal_resources/1914/



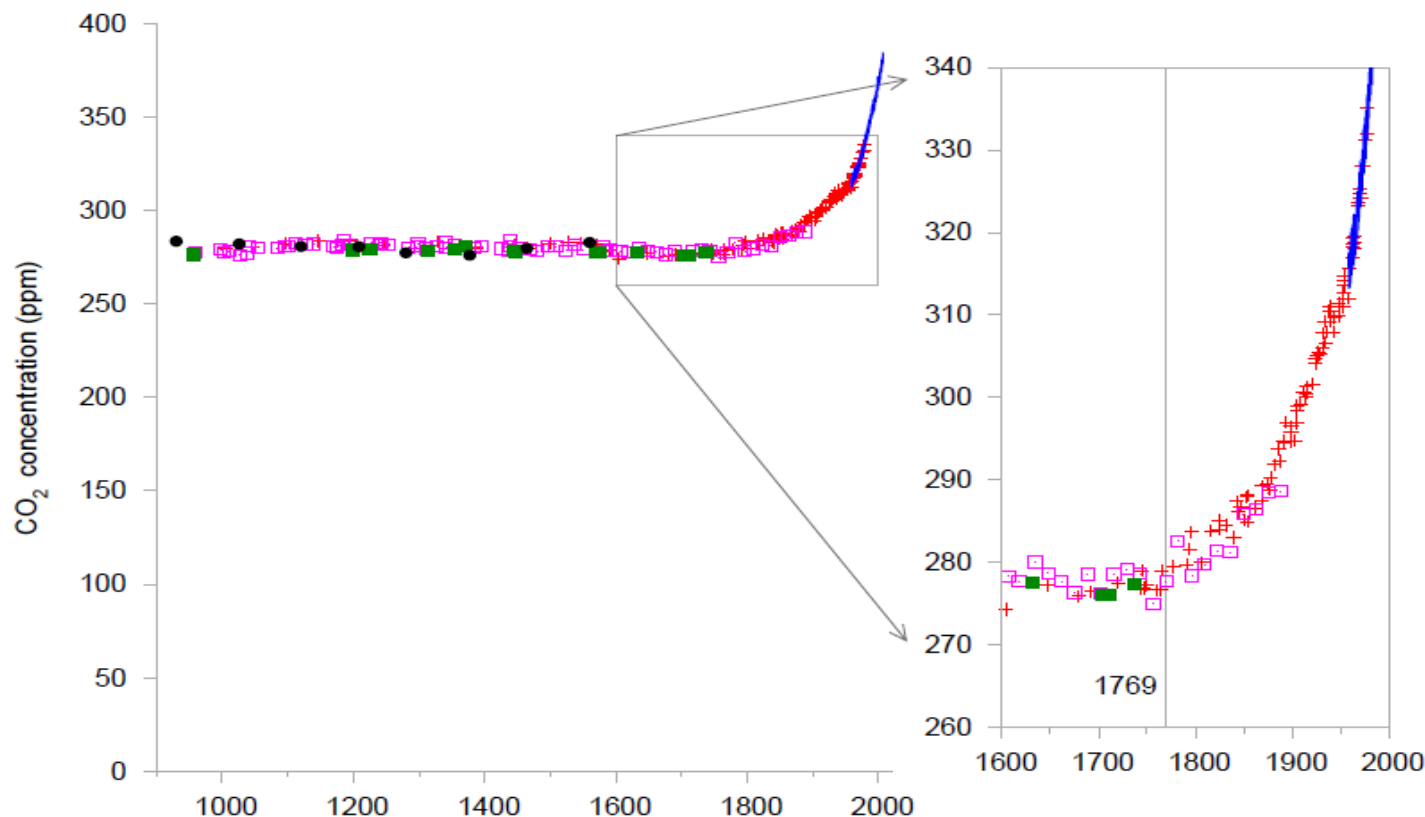
Fonte: "State of the global climate 2020", Organização Meteorológica Mundial.

Energia e sustentabilidade



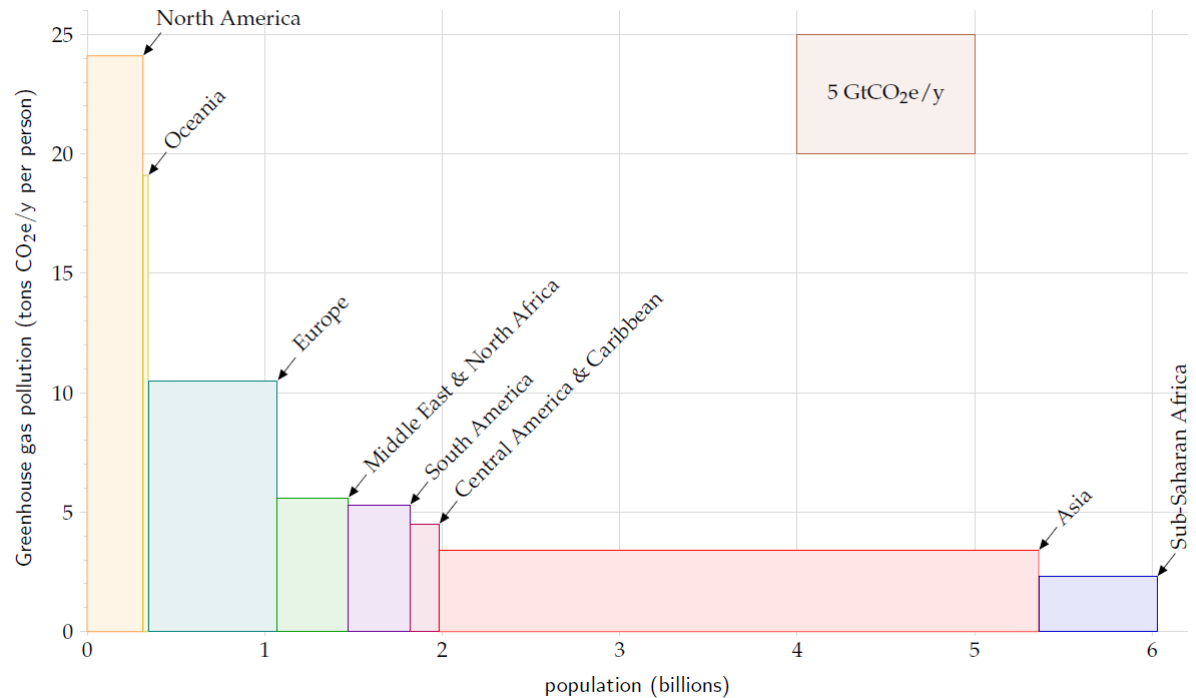
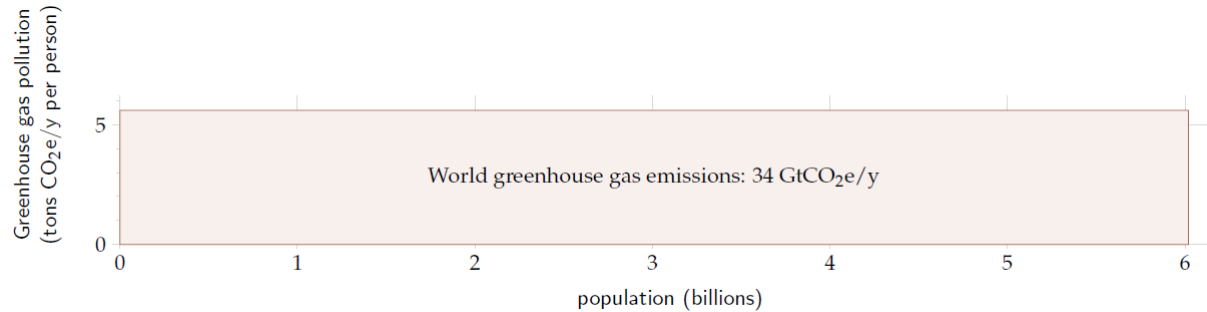
Energia e sustentabilidade

Carbon dioxide (CO_2) concentrations (in parts per million) for the last 1100 years, measured from air trapped in ice cores (up to 1977) and directly in Hawaii (from 1958 onwards).



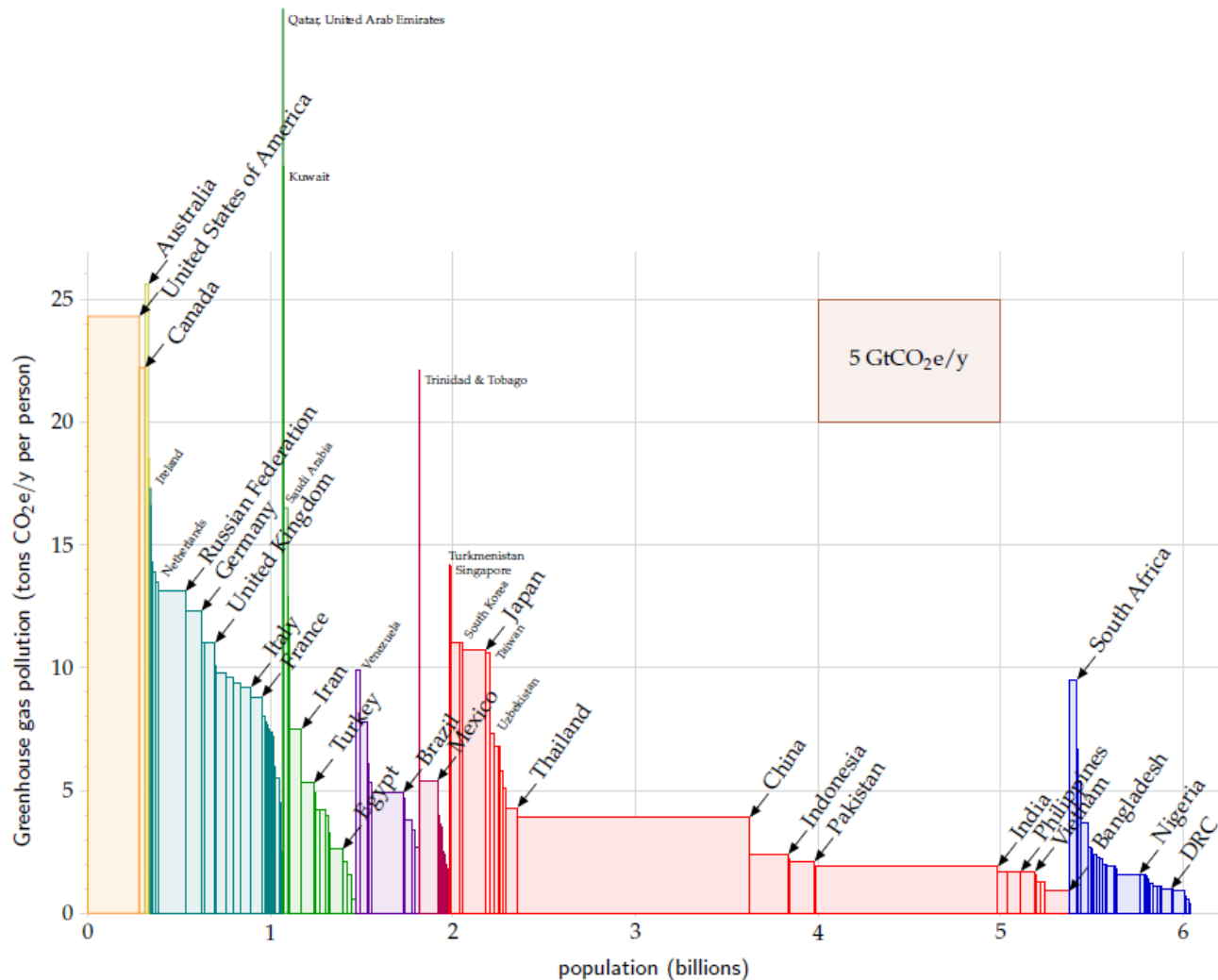
Energia e sustentabilidade

Emissões de CO₂ per capita



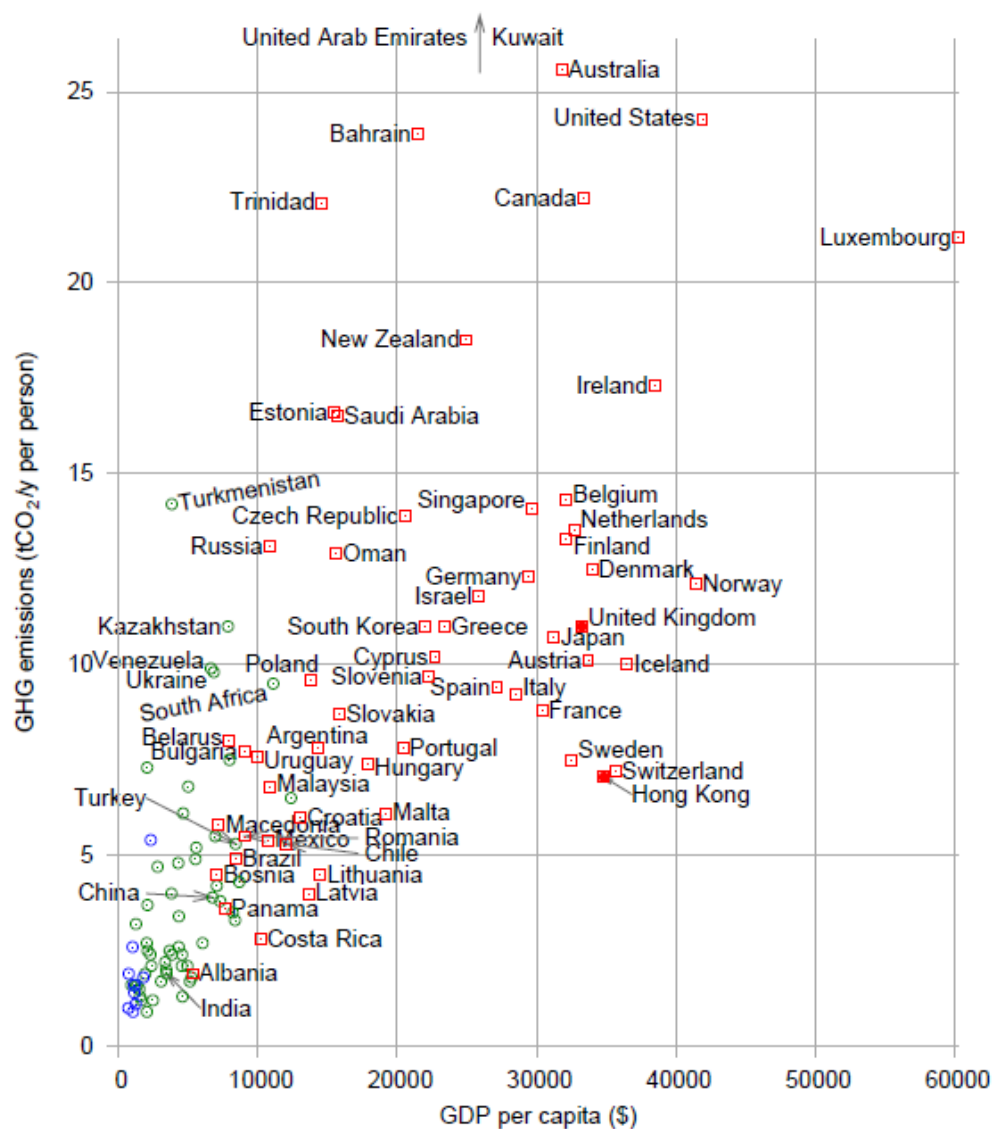
Energia e sustentabilidade

Emissões de CO₂ per capita



Energia e sustentabilidade

Emissões de CO₂ per capita



Energia e sustentabilidade

Emissões de CO₂ por atividade/setor

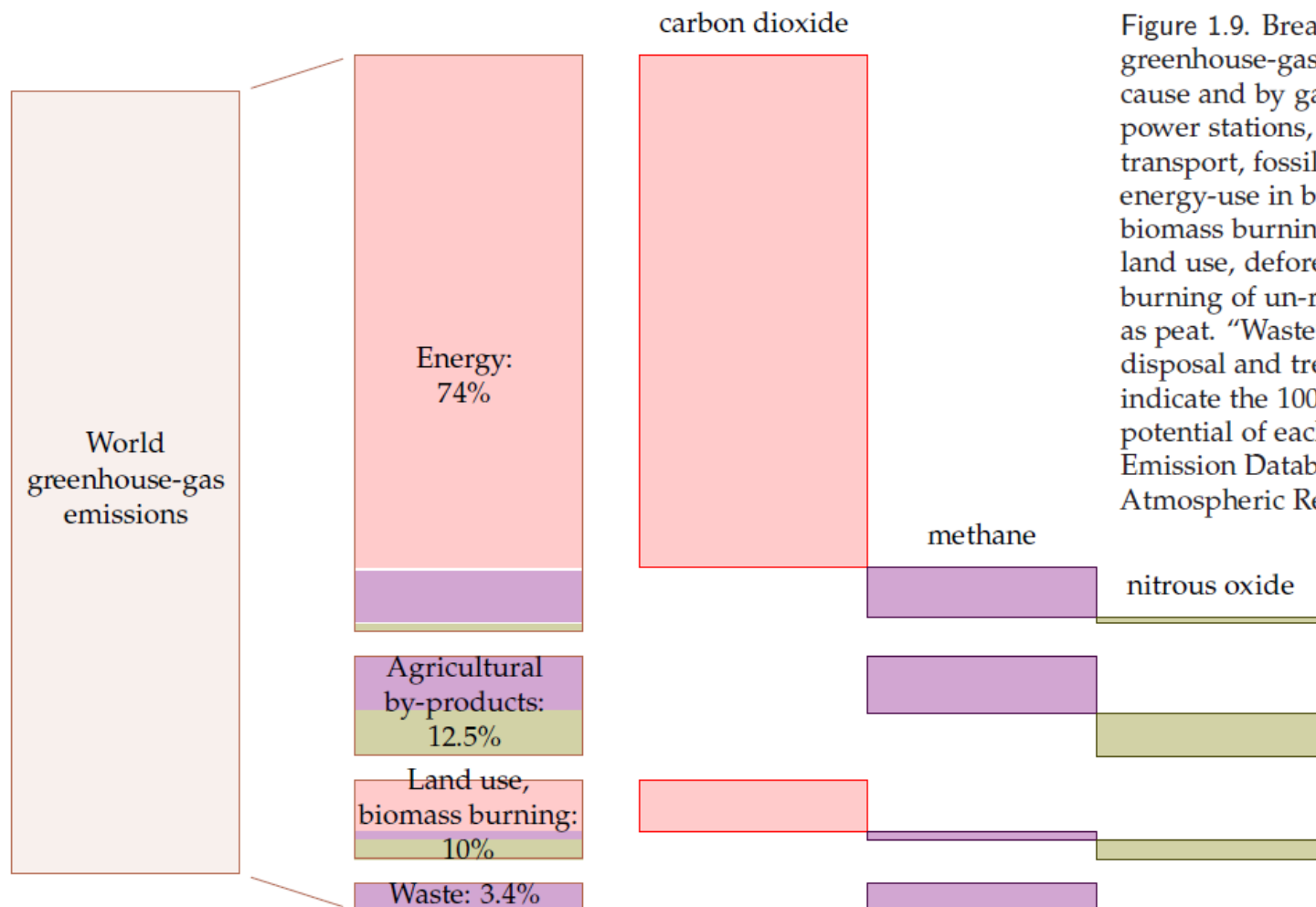
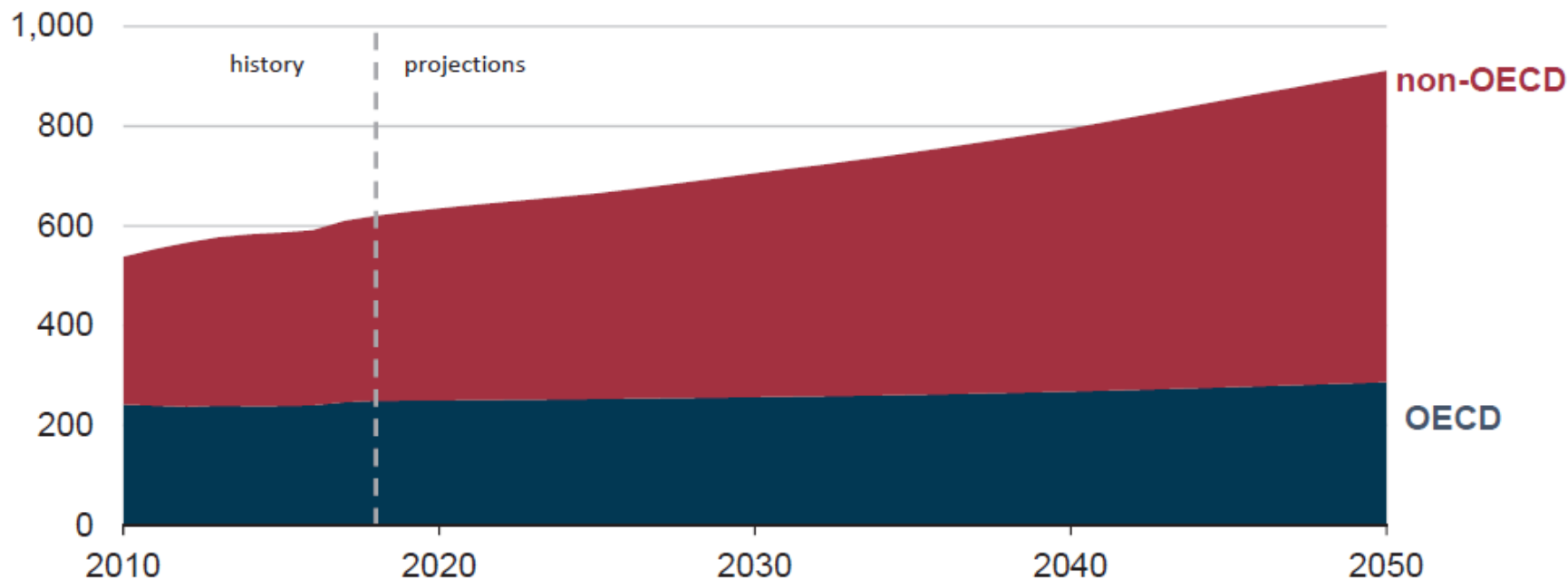


Figure 1.9. Breakdown of world greenhouse-gas emissions (2000) by cause and by gas. “Energy” includes power stations, industrial processes, transport, fossil fuel processing, and energy-use in buildings. “Land use, biomass burning” means changes in land use, deforestation, and the burning of un-renewed biomass such as peat. “Waste” includes waste disposal and treatment. The sizes indicate the 100-year global warming potential of each source. Source: Emission Database for Global Atmospheric Research.

Energia e sustentabilidade

Estima-se que o consumo de energia aumente 50% entre 2018 e 2050!!

World energy consumption
quadrillion British thermal units

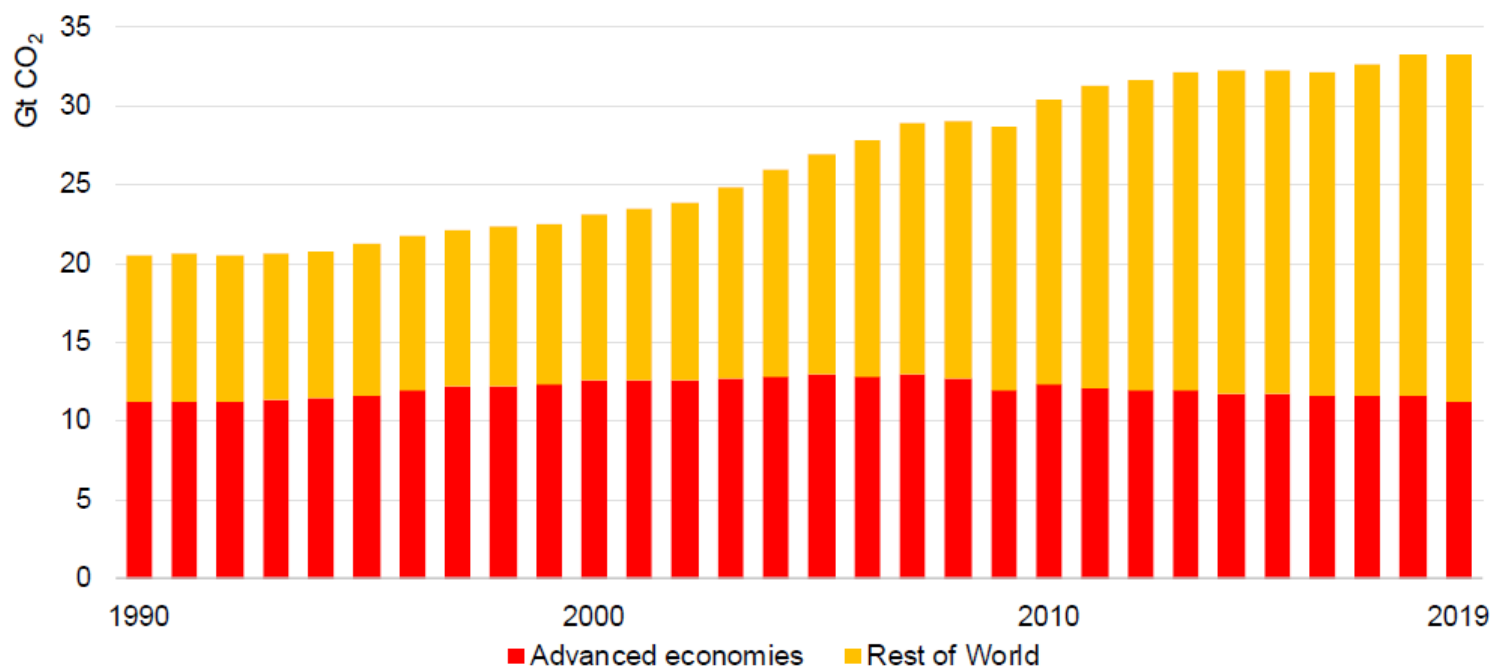


Nota: 1 BUT ~ 1055 J

Energia e sustentabilidade

Emissões de CO₂ associadas à produção de energia

Energy-related CO₂ emissions, 1990-2019



IEA 2020. All rights reserved.

Global CO₂ emissions flattened in 2019 at around 33.2 Gt, after two years of increases. Lower coal-fired generation in advanced economies and rising output from lower-carbon sources underpinned the decline.

Energia e sustentabilidade

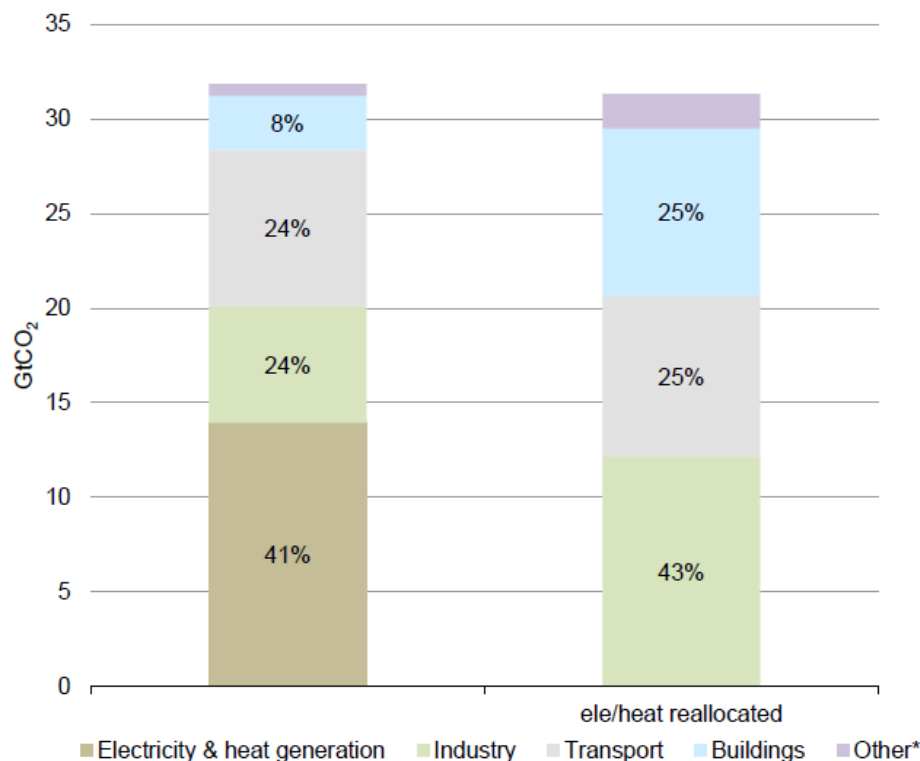
- Para lá daquilo que o consumo de energia representa em termos de utilização das reservas fósseis não renováveis, a face menos visível e com mais impacto ambiental do consumo de energia, tem a ver com as emissões de carbono.
- A maioria destas emissões são geradas durante a queima de carvão e gás para produção de eletricidade nas centrais termoelétricas, no setor dos transportes e no setor industrial.



Energia e sustentabilidade

Emissões de CO₂ geradas pela combustão de combustível -por setor de atividade

Global CO₂ emissions by sector, 2018



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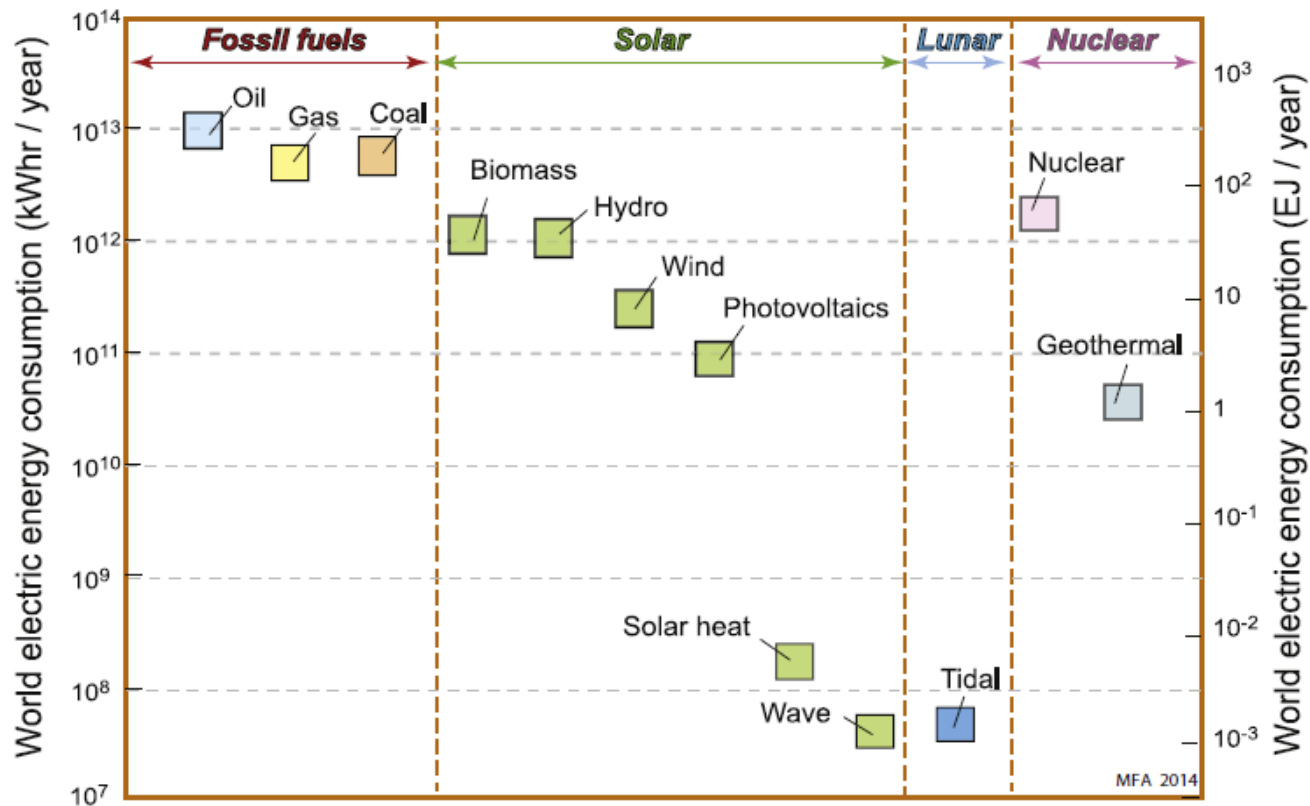
* Other includes agriculture/forestry, fishing and non-specified final energy consumption.

Power generation, together with transport, accounted for over two thirds of total emissions in 2018; industry was the largest final emitting sector.

Energia e sustentabilidade

Consumo mundial de energia por recurso

2014



As unidades da direita são em exajoule (10¹⁸) e as da esquerda, mais familiares, em kWh.

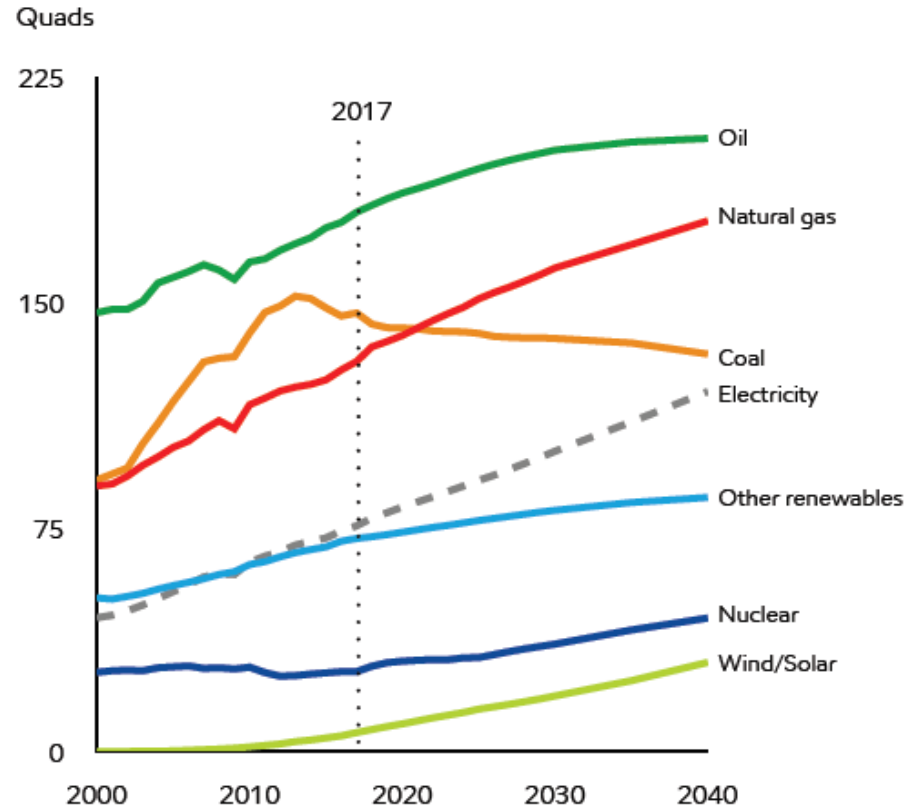
$$1 \text{ J} = 2,778 \times 10^{-7} \text{ kWh}$$

Energia e sustentabilidade

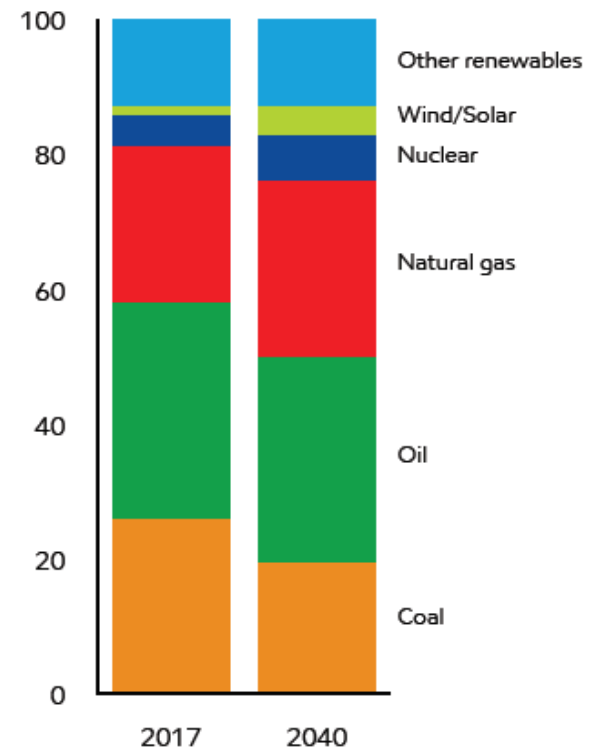
Consumo mundial de energia por recurso

2017

Global energy mix shifts to lower-carbon fuels



Percent of primary energy (%)

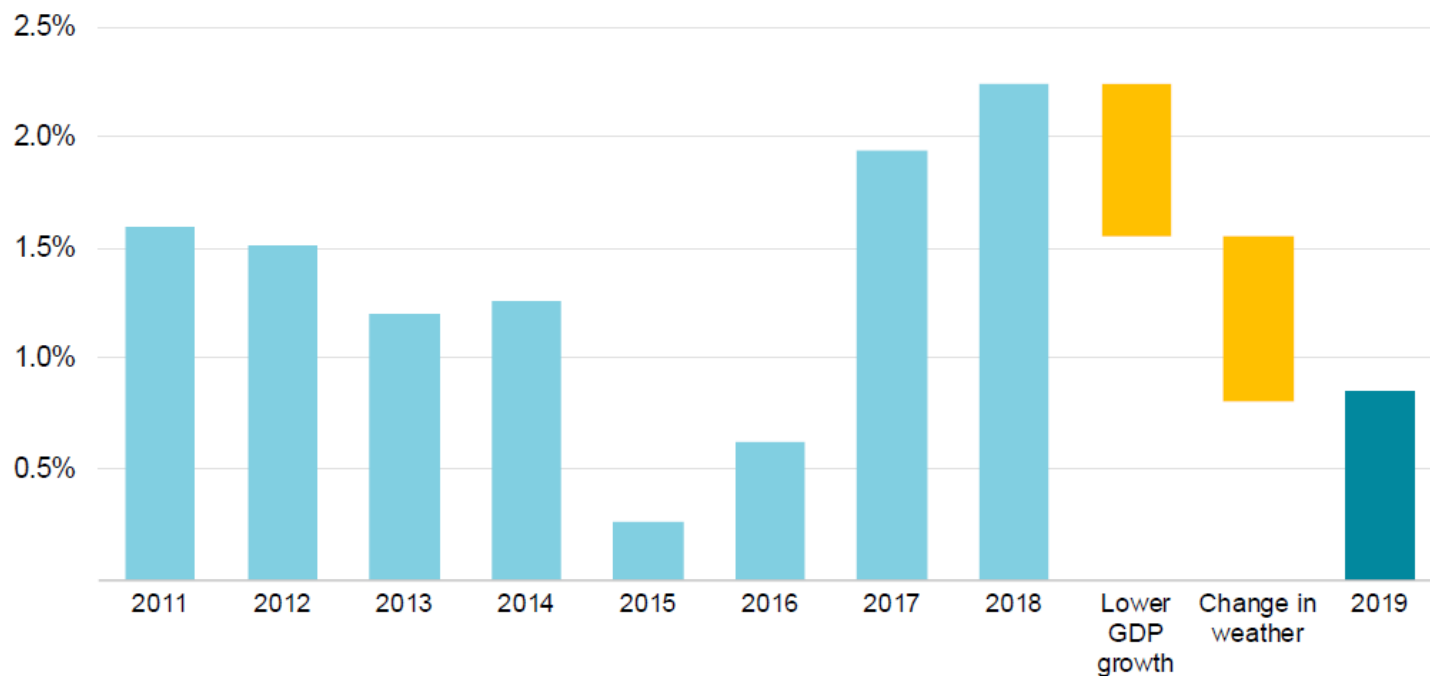


Nota: 1 Quad = $1,055 \times 10^{18}$ J

Energia e sustentabilidade

Consumo mundial de energia, variação

World energy demand growth rate, 2011-19



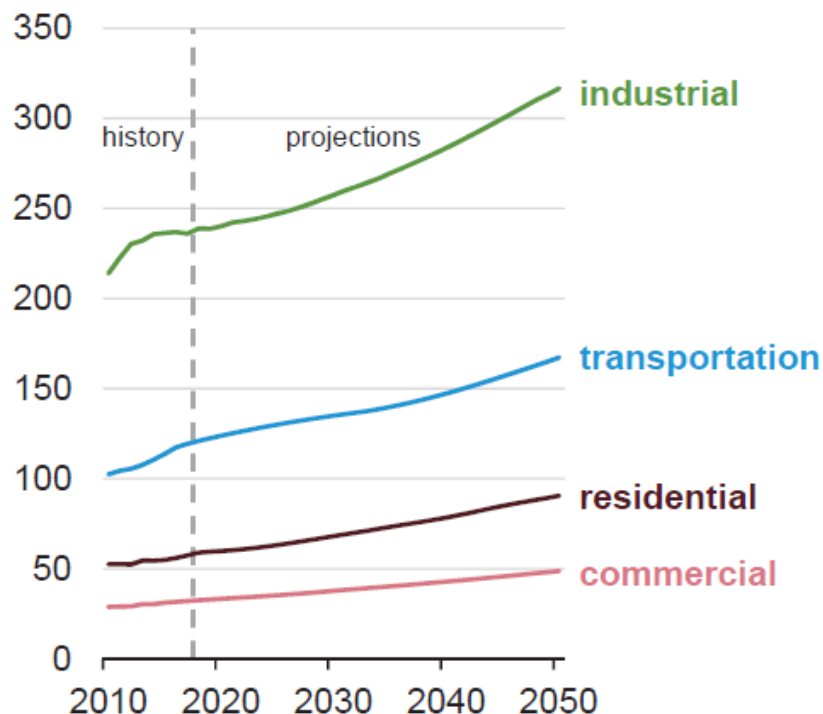
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Global energy demand grew by 0.9% in 2019, a sharp decline from 2018, due mostly to slower economic growth and changes in weather.

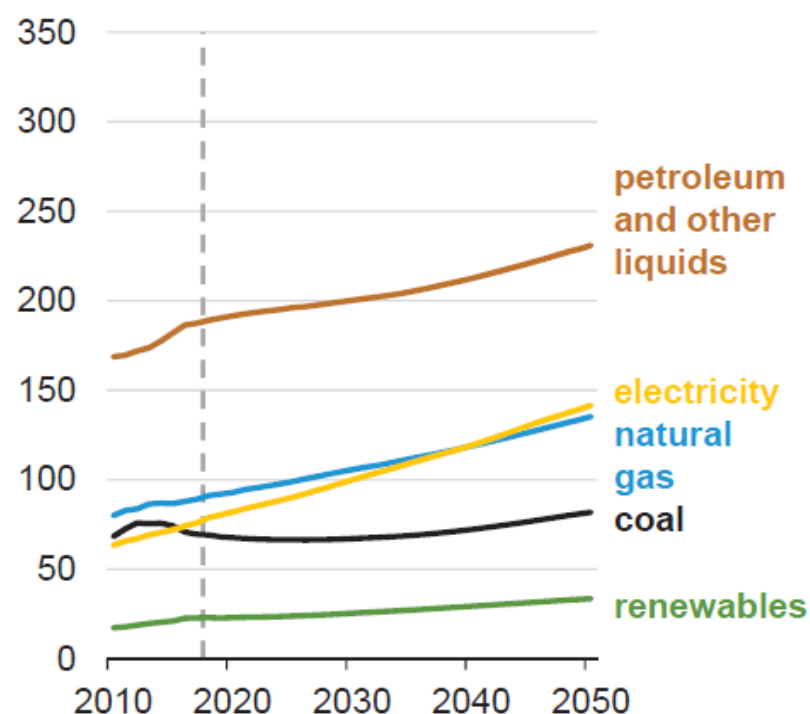
Energia e sustentabilidade

Consumo mundial de energia por recurso e por setor

End-use energy consumption by sector, world
quadrillion British thermal units



End-use energy consumption by fuel, world
quadrillion British thermal units

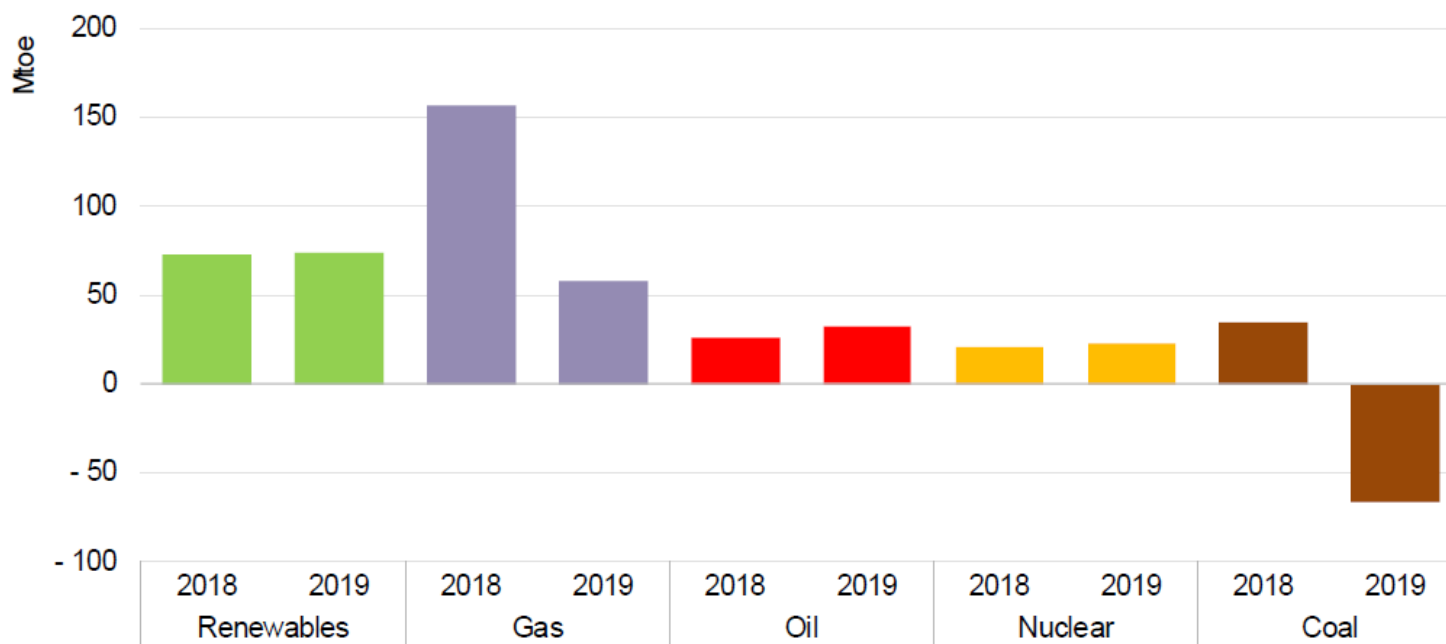


Nota: 1 BUT ~ 1055 J

Energia e sustentabilidade

Variação na procura de energia por tipologia

Average annual change in energy demand by fuel, 2018 and 2019



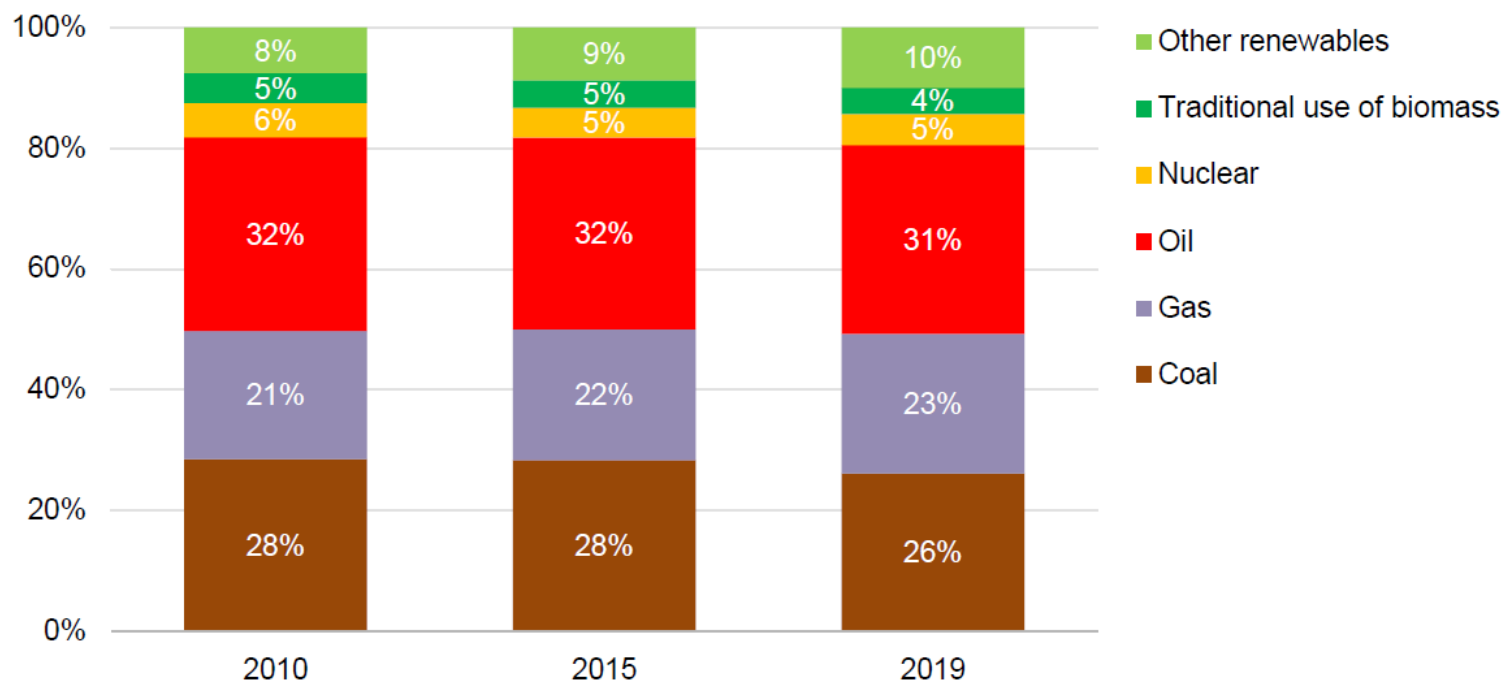
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Renewables led the growth in energy demand, followed by gas, while coal declined steeply.

Energia e sustentabilidade

Variação na procura de energia por tipologia

Share of total primary energy demand by fuel, 2010-19



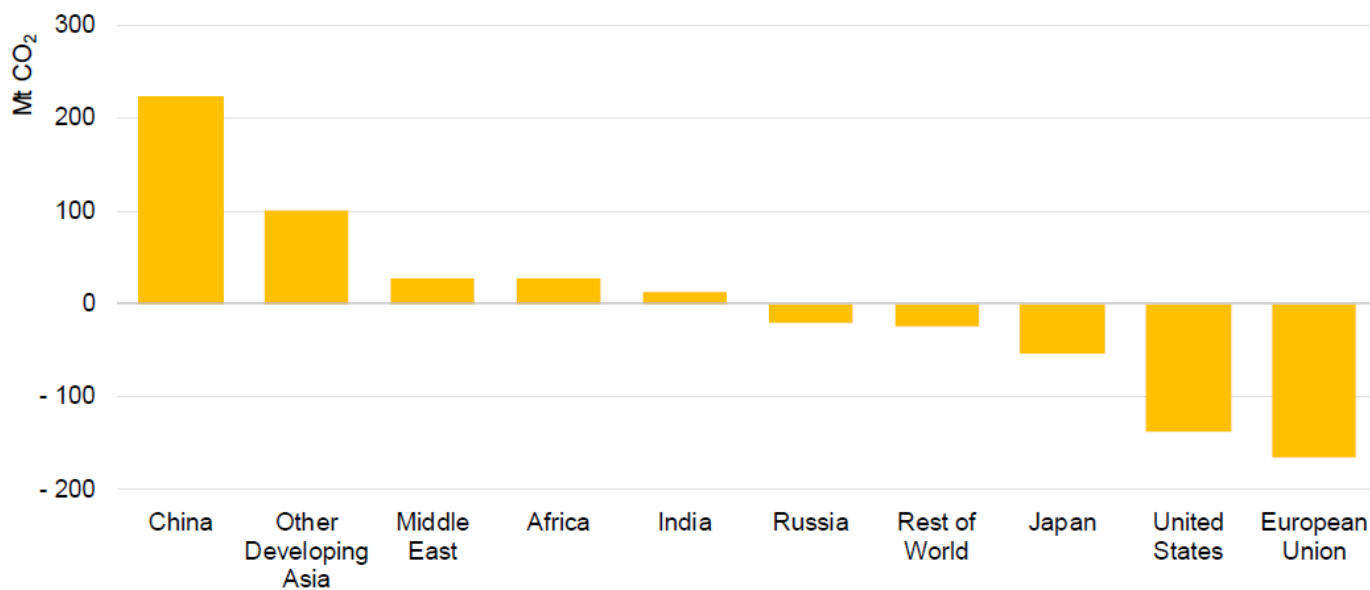
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The share of natural gas in total primary energy demand continued to increase, reaching 23%, while modern renewables passed 10% in 2019.

Energia e sustentabilidade

Emissões de CO₂ associadas à produção de energia

Change in energy-related CO₂ emissions by region, 2018-19



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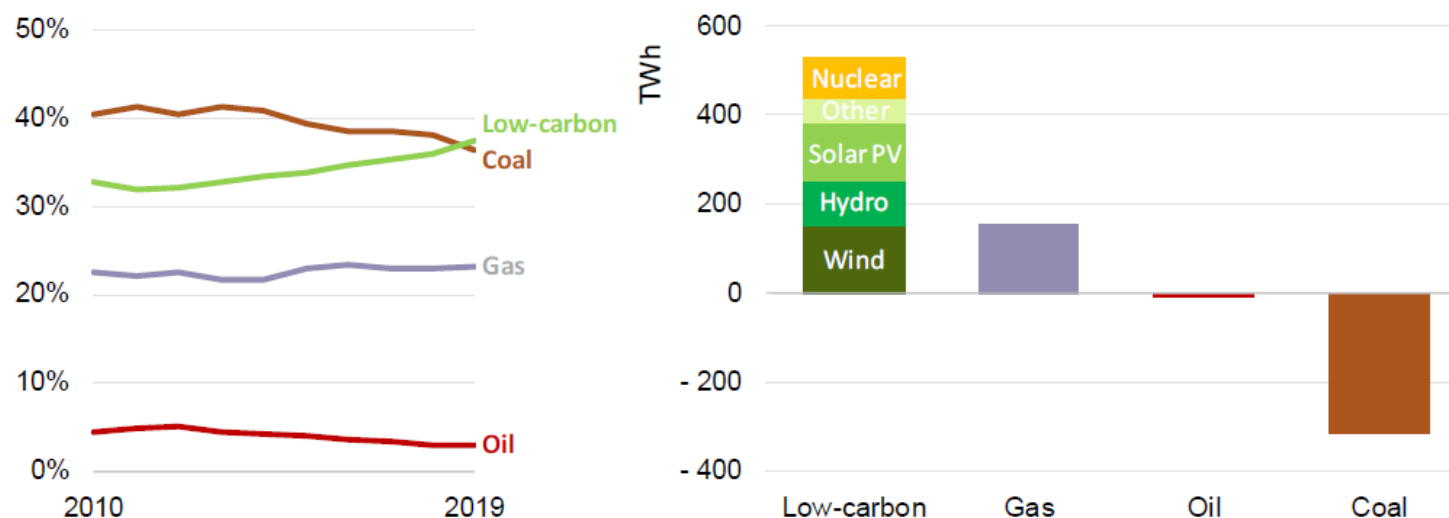
Note: China = the People's Republic of China; Russia = the Russian Federation.

Emissions declined across advanced economies, led by lower emissions from the power sector. In the rest of the world, especially in India, emissions grew more slowly because of lower economic growth.

Energia e sustentabilidade

Produção de energia elétrica – mudança de paradigma?

Global power mix over time and year-on-year change in electricity generation by source in 2019



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2019 marked a key turning point in the global power mix, with electricity generation from low carbon sources exceeding that of coal for the first time.

Energia e sustentabilidade

Produção de energia elétrica – mudança de paradigma?

World summary

	Energy Demand (Mtoe)	Growth rate (%)	Shares (%)	
	2019	2018-19	2000	2019
Total Primary Energy Demand	14 385	0.9%	100%	100%
Coal	3 762	-1.7%	23%	26%
Oil	4 512	0.7%	37%	31%
Gas	3 320	1.8%	21%	23%
Nuclear	732	3.3%	7%	5%
Hydro	373	2.4%	2%	3%
Biomass and waste	1 362	2.2%	10%	9%
Other renewables	326	12.2%	1%	2%

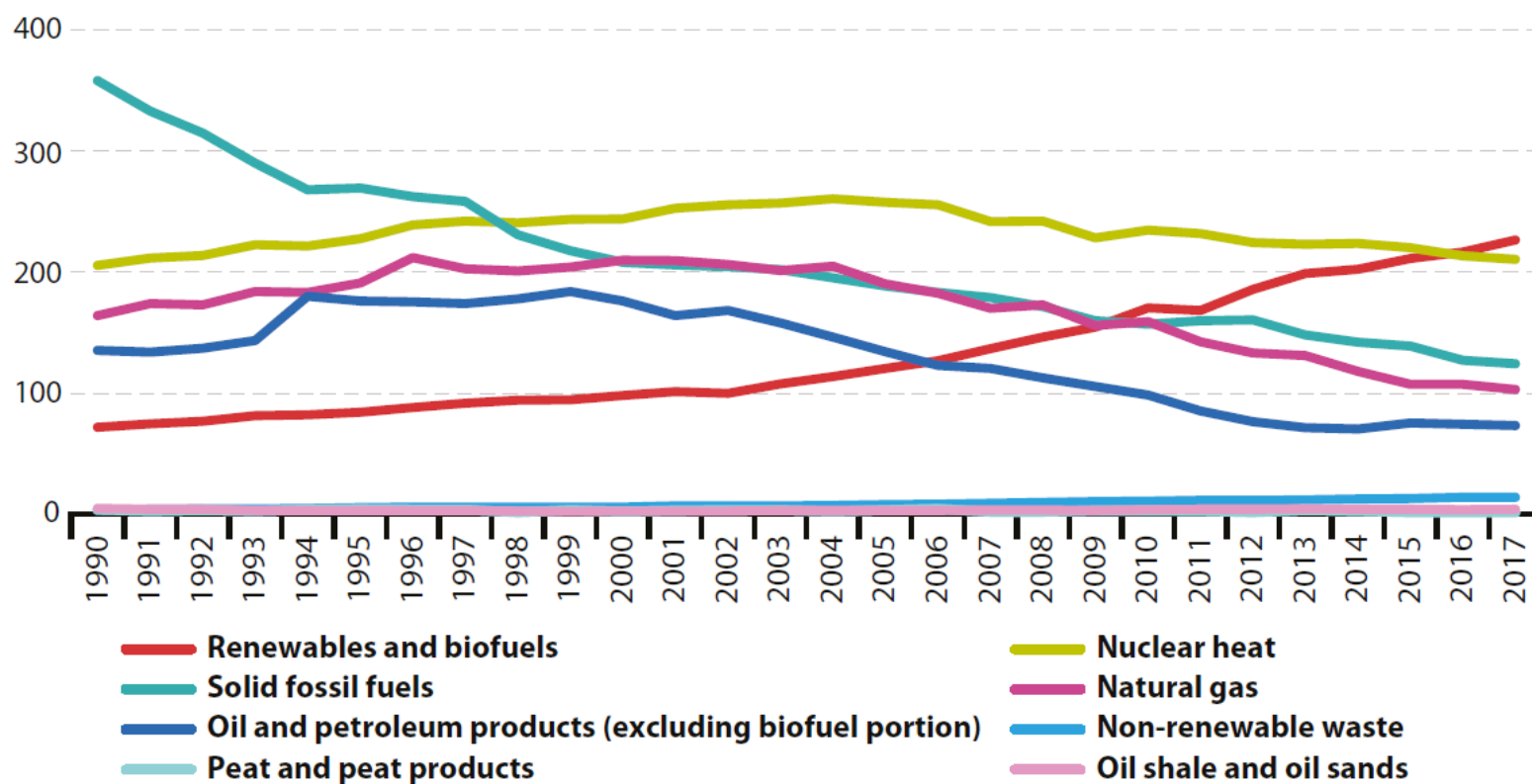
	Electricity Generation (TWh)	Growth rate (%)	Shares (%)	
	2019	2018-19	2000	2019
Total Generation	26 951	1.3%	100%	100%
Coal	9 819	-3.1%	39%	36%
Oil	789	-1.4%	8%	3%
Gas	6 246	2.5%	18%	23%
Nuclear	2 806	3.3%	17%	10%
Hydro	4 333	2.4%	17%	16%
Biomass and waste	688	7.8%	1%	3%
Wind	1 405	11.9%	0%	5%
Solar photovoltaics	720	22.3%	0%	3%
Other renewables	144	5.3%	0%	1%

Energia e sustentabilidade

Produção de energia – ponto da situação na EU28

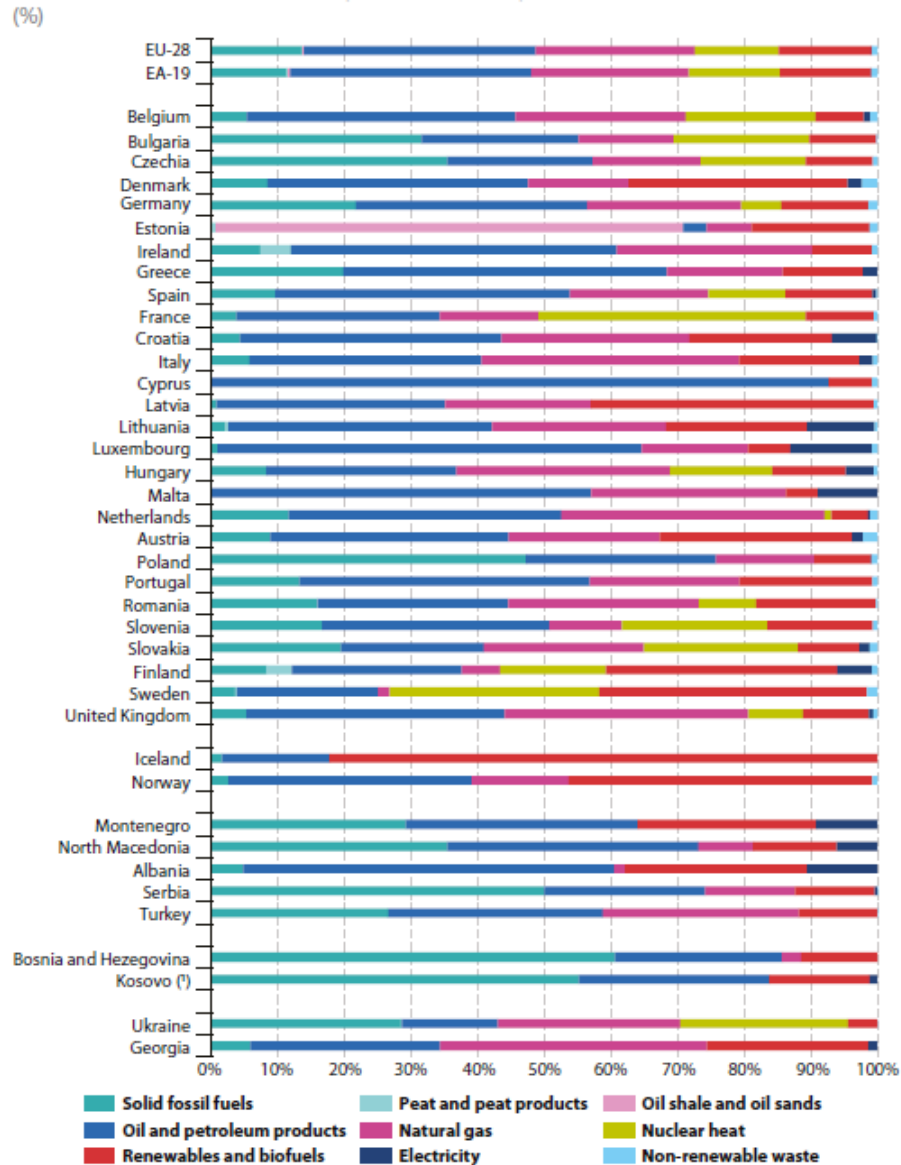
Figure 1.1.1: Primary energy production by fuel, EU-28, 1990-2017

(Mtoe)



Source: Eurostat (online data code: [nrg_bal_c](#))

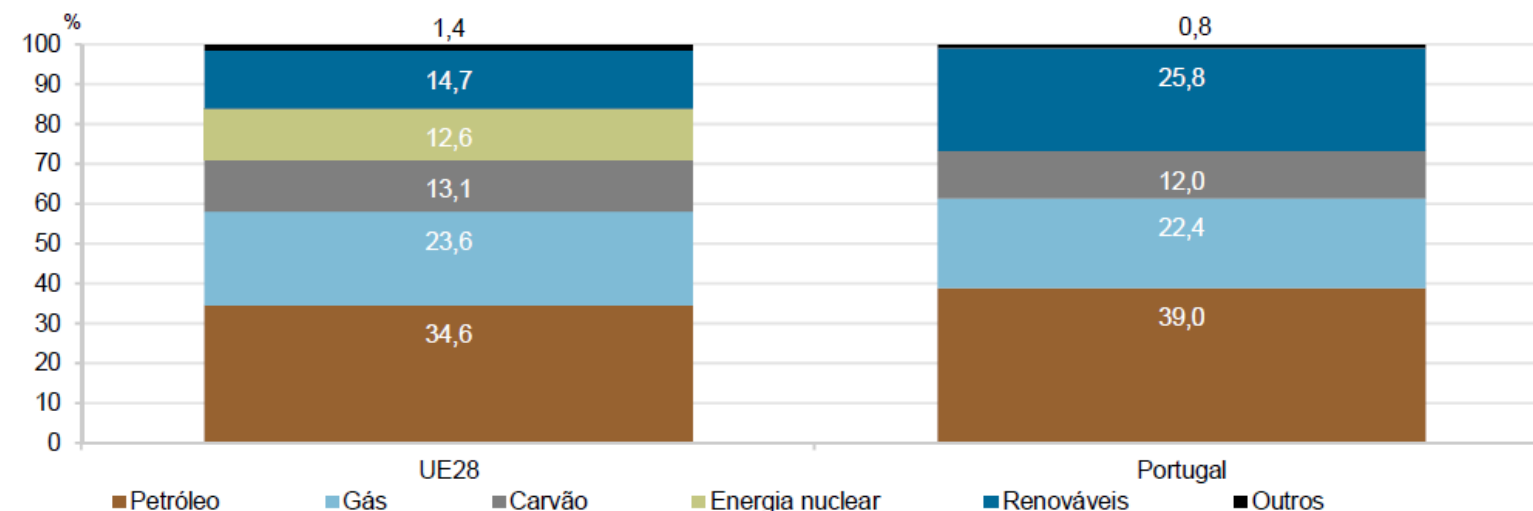
Consumo de energia por combustível – ponto da situação na EU28 (2017)



Fonte: “Energy, transport and environment statistics”, Eurostat (2019).

Contexto em Portugal

Figura 6.2 >> Consumo de energia primária por fonte energética - UE28 e Portugal (2018)

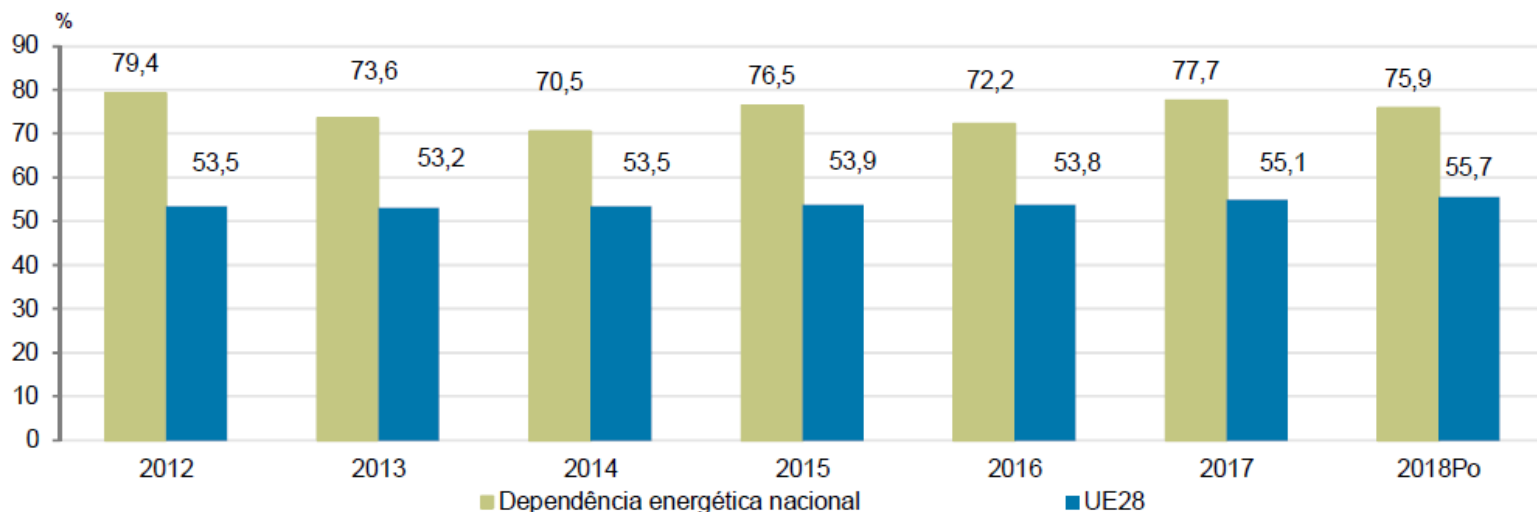


Fonte: DGEG

De acordo com os dados de 2018 relativos ao consumo de energia primária por fonte energética na UE28 e em Portugal, verifica-se que Portugal tem uma maior dependência do petróleo, 39,0% face a 34,6% da UE28, sendo no entanto evidente a crescente oferta energética proveniente de fontes renováveis (25,8%), claramente superior à média da UE28 (14,7%). Realça-se ainda a importância que a energia nuclear tem como fonte de energia primária na UE28, representando 12,6% do total de energia primária consumida em 2018.

Contexto em Portugal

Figura 6.3 >> Dependência energética nacional

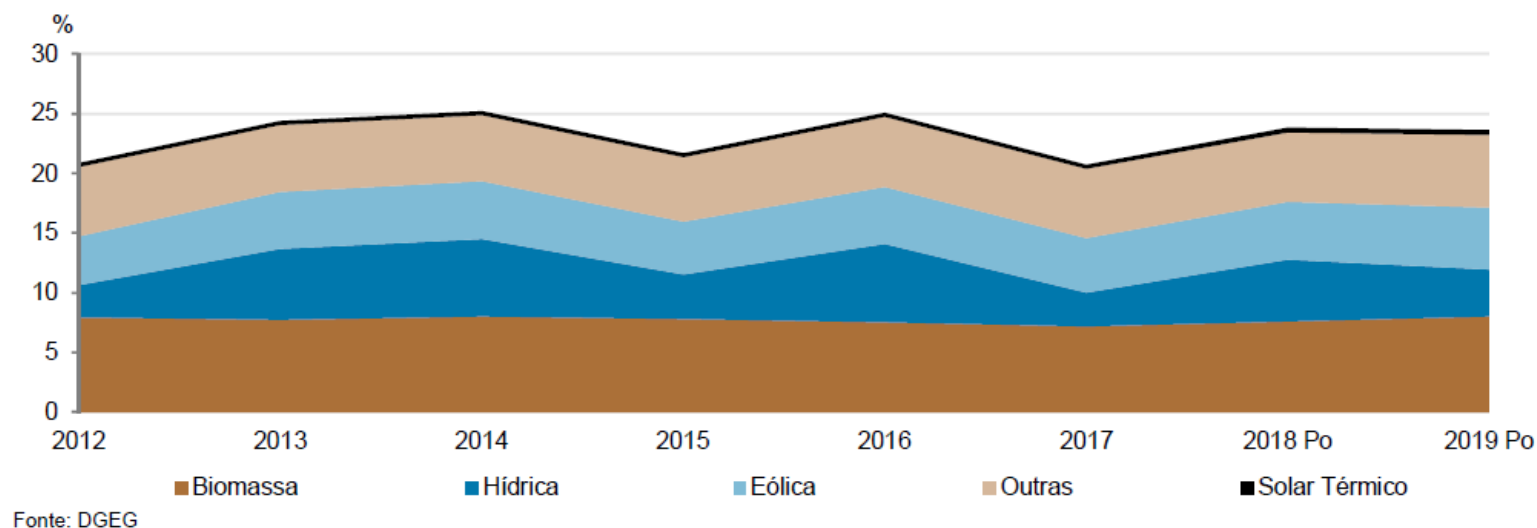


Fonte: DGEG

Em 2018, 75,9% da energia primária consumida em Portugal foi importada. Face a 2017, a dependência energética nacional diminuiu 1,8 p.p., o que se deveu sobretudo ao decréscimo das importações de carvão.

Contexto em Portugal

Figura 6.10 >> Proporção de fontes renováveis no consumo de energia primária

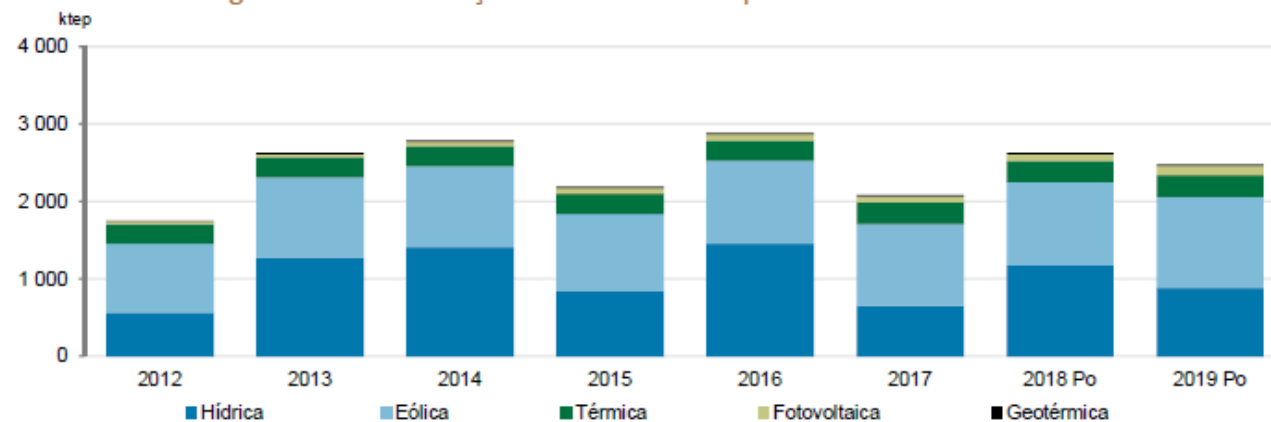


A contribuição das fontes de energia renováveis para o consumo de energia primária foi 23,7% em 2019 (-0,2 p.p. face a 2018). Esta diminuição resultou da menor contribuição da energia hídrica para o total das energias renováveis no consumo primário em 2019 (3,9%, quando em 2018 foi de 5,2%).

A biomassa (lenhas e resíduos florestais, biogás e biodiesel) continuou a ser, em 2018, a fonte de energia renovável com maior contribuição para o consumo primário com 8,0% (7,6% em 2018).

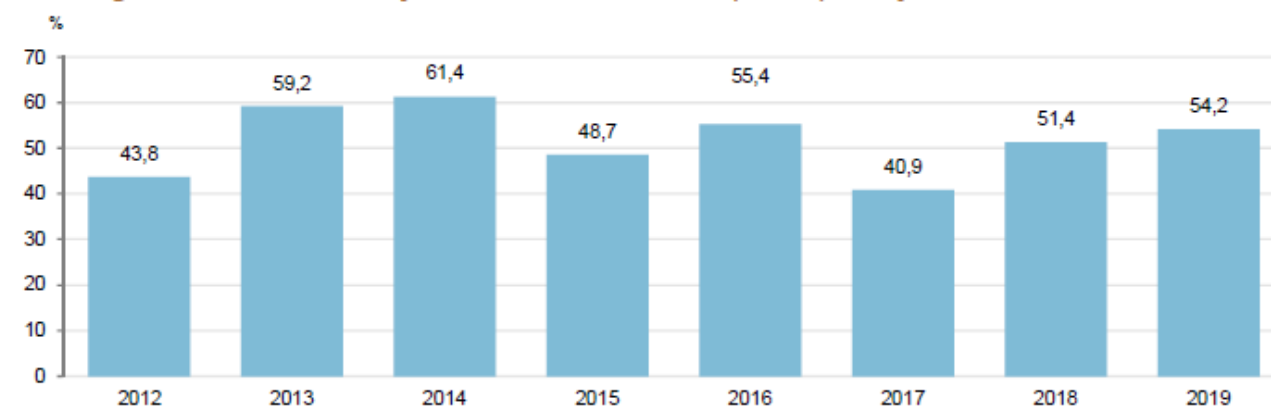
Contexto em Portugal

Figura 6.11 >> Produção de eletricidade a partir de fontes renováveis



Fonte: DGEG

Figura 6.12 >> Contribuição das fontes renováveis para a produção total de eletricidade

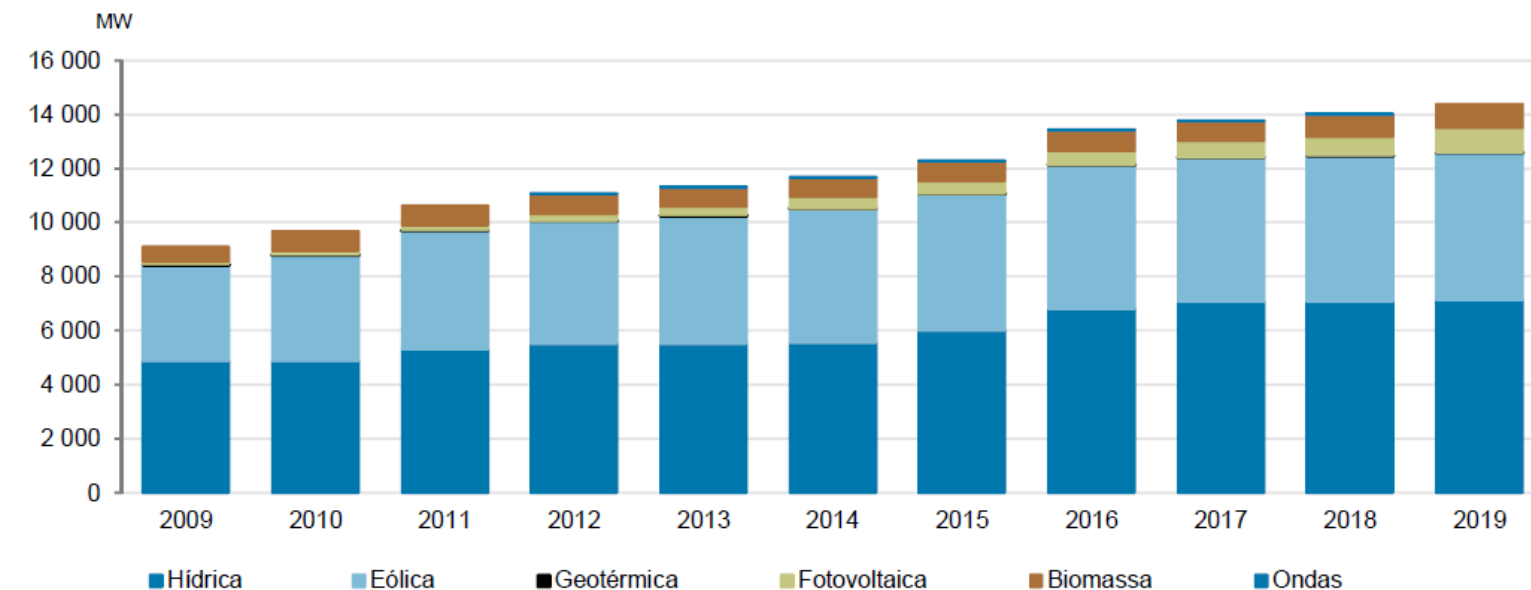


Fonte: DGEG

Contexto em Portugal

Em termos do coeficiente/fonte para o total da produção de eletricidade a partir das fontes renováveis, em 2019 a componente eólica representou 47,4% (41,2% em 2018), a hídrica 35,5% (44,5% em 2018) e a térmica 11,7% (10,3% em 2018).

Figura 6.13 >> Capacidade instalada de energias renováveis

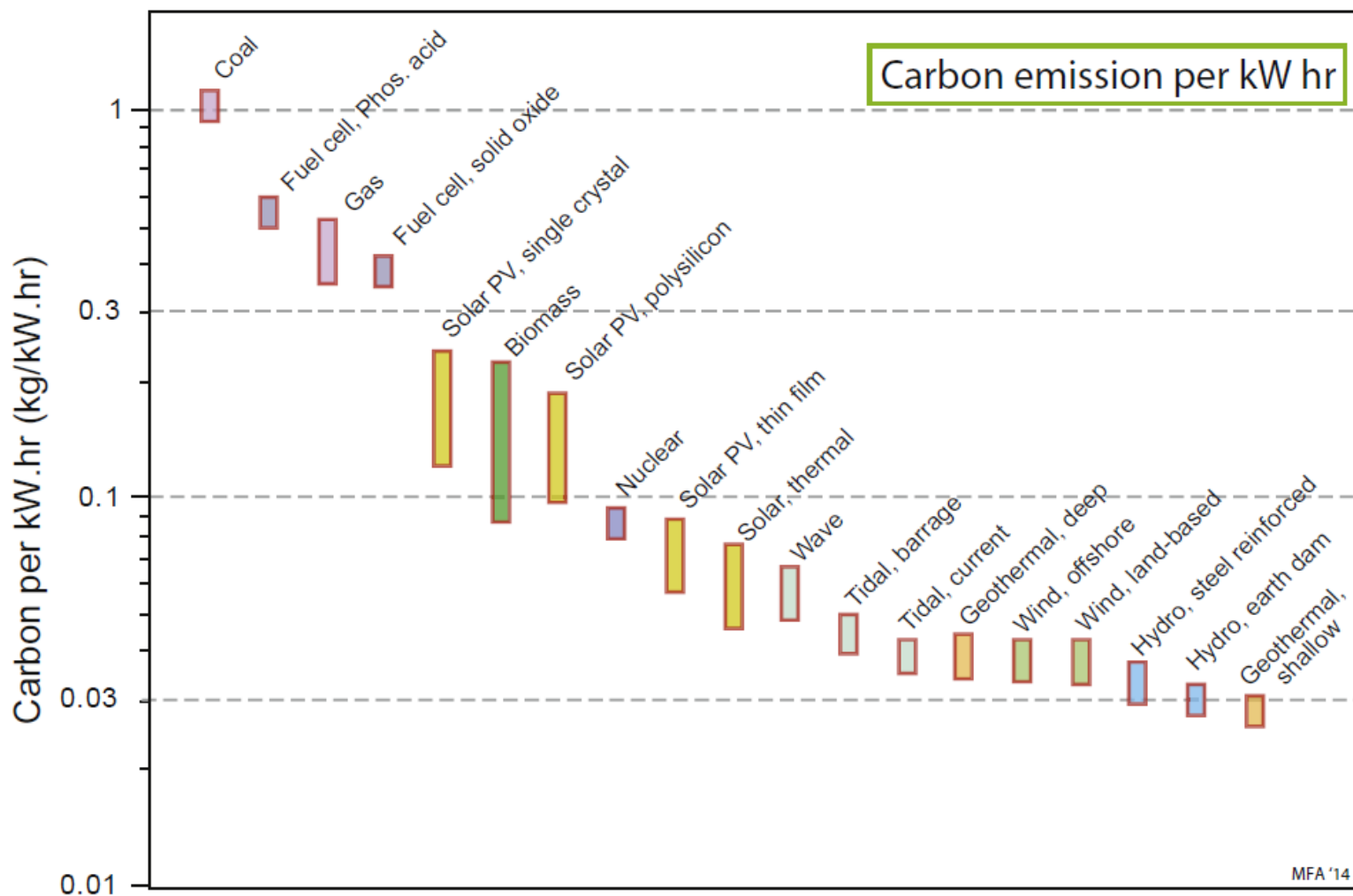


Fonte: DGE

Em 2019, a potência total instalada de energias renováveis foi 14 402 MW, a qual apresentou desde 2009 um crescimento contínuo a uma taxa média anual de 4,3%, em resultado essencialmente do aumento de 2 247 MW de potência instalada de energia hídrica e de 1 893 MW de energia eólica.

Energia e sustentabilidade

Pegada de carbono de sistemas de geração de energia



Energia e sustentabilidade

Pegada de carbono de sistemas de geração de energia

Fuel Type	kg OE ^a	MJ/liter	MJ/kg	CO _{2,eq} kg/liter	CO _{2,eq} kg/kg	CO _{2,eq} kg/MJ
Coal, lignite	0.45	-	18–22	-	1.6	0.080
Coal, anthracite	0.72	-	30–34	-	2.9	0.088
Crude oil	1.0	38	44	3.1	3.0	0.070
Diesel	1.0	38	44	3.1	3.2	0.071
Gasoline	1.05	35	45	2.9	2.89	0.065
Kerosene	1.0	35	46	3.0	3.0	0.068
Ethanol	0.71	23	31	2.8	2.6	0.083
LNG	1.2	25	55	3.03	3.03	0.055
Hydrogen	2.7	8.5	120	0	0	0

^akilograms oil equivalent (the kg of oil with the same energy content).

Energia e sustentabilidade

Intensidade energética e pegada de carbono de diferentes sistemas de geração de energia

Power System	Capital Intensity (k\$/kW _p) ^a	Area Intensity (m ² /kW _p)	Material Intensity (kg/kW _p)	Construction Energy Intensity (MJ/kW _p)	Construction Carbon Intensity (kg/kW _p)	Capacity Factor (%)
Conventional gas	0.6–1.5	1–4	605–1080	1,730–2,710	100–200	75–85
Conventional, coal	2.5–4.5	1.5–3.5	700–1600	3,580–9,570	100–700	75–85
Phosphoric acid fuel cell	3–4.5	0.1–0.5	80–120	5,000–10,000	600–1000	>95
Solid oxide fuel cell	7–8	0.3–1	50–100	2,000–6,000	200–400	>95
Nuclear - fission	3.5–6.4	1–3	170–625	2,000–4,300	105–330	75–95
Wind, land-based	1.0–2.4	150–400	500–2,000	3,500–6,000	240–600	17–25
Wind, off-shore	1.6–3	100–300	300–900	5,000–10,000	480–1,000	30–40
Solar PV, single crystal	4–12	30–70	800–1,700	30,000–60,000	2,000–4,000	8–12 ^b
Solar PV, poly-silicon	3–6 ^b	50–80 ^b	1,000–2,000	20,000–40,000	1,500–3,000	8–12 ^b
Solar PV, thin-film	2–5	50–100	1,500–3,000	10,000–20,000	550–1,000	8–12 ^b
Solar thermal	3.9–8	20–100	650–3,500	19,000–40,000	1,500–3,500	20–35 ^c
Hydro-earth dam	1–5	200–600	15,000–100,000	7260–15,000	630–1,200	45–65

(Continued)

Energia e sustentabilidade

Intensidade energética e pegada de carbono de diferentes sistemas de geração de energia

Power System	Capital Intensity (k\$/kW _p) ^a	Area Intensity (m ² /kW _p)	Material Intensity (kg/kW _p)	Construction Energy Intensity (MJ/kW _p)	Construction Carbon Intensity (kg/kW _p)	Capacity Factor (%)
Hydro- steel reinforced	1–5	120–500	8,000–40,000	30,000–66,000	1,000–4,000	50–70
Wave	1.2–4.4	42–100	1,000–2,000	22,950–31,540	1670–2070	25–40
Tidal-current	10–15	150–200	350–650	12,000–18,000	800–1130	35–50
Tidal-barrage	1.6–2.5	200–300	5,000–50,000	30,000–45,000	2,400–3,520	20–30
Geothermal-shallow	1.15–2	1–3	61–500	7,000–13,500	160–250	75–95
Geothermal-deep	2–3.9	1–3	400–1200	20,000–40,700	1,700–3,900	75–95
Biomass-dedicated	2.3–3.6	10,000–33,000	500–922	5,000–19,800	600–1800	75–95

Limites do desenvolvimento/crescimento de renováveis

Energia eólica



How much wind power could we plausibly generate?

power per person = wind power per unit area $\times$ area per person.

Chapter B (p263) explains how to estimate the power per unit area of a wind farm in the UK. If the typical windspeed is 6 m/s (13 miles per hour, or 22 km/h), the power per unit area of wind farm is about 2 W/m^2 .

$$250 \text{ pessoas/km}^2 = 4000 \text{ m}^2/\text{pessoa}$$

$$2 \text{ W/m}^2 \times 4000 \text{ m}^2/\text{pessoa} = 8000 \text{ W/pessoa} = 200 \text{ kWh/d/pessoa}$$

Cenário:

utilizar 10% da área do país

CONSUMPTION PRODUCTION

Car:
40 kWh/d

Wind:
20 kWh/d



Limites do desenvolvimento/crescimento de renováveis

Energia eólica em Portugal



$$100 \text{ pessoas/km}^2 = 10000 \text{ m}^2/\text{pessoa}$$

$$2 \text{ W/m}^2 \times 10000 \text{ m}^2/\text{pessoa} = 20000 \text{ W/pessoa} = 480 \text{ kWh/d.pessoa}$$

Cenário: utilizar 10% da área do país = 48 kWh/d.pessoa

Consumo eletricidade em Portugal (2018, Pordata) = 48893842496 KWh
(Dividindo por 10 Milhões e 365 dias = 13,4 kWh/d.p)

Considerando intermitência da geração, tipicamente de 20%, o contributo das eólicas com 10% da área coberta seria de 9,6 kWh/d.p.

A estatística de produção de eletricidade em PT (ainda Pordata, 2018), indica que a componente eólica foi de 12617 GWh (cerca de 3,5 kWh/d.p)

Menor Eficiência global ou 3-4% da área coberta !!!



Energia eólica – DESAFIOS/LIMITAÇÕES

- **Uso de Materiais Críticos (Nd e B nos ímãs permanentes)**
- **Mobilização de terra (ou mar, se offshore) e impactos (visuais, ruído)**
- **Fim de vida de alguns componentes (ao fim de cerca de 20 anos): compostos de fibra de vidro das pás de difícil reciclagem**
- **Largas regiões com pouco vento**

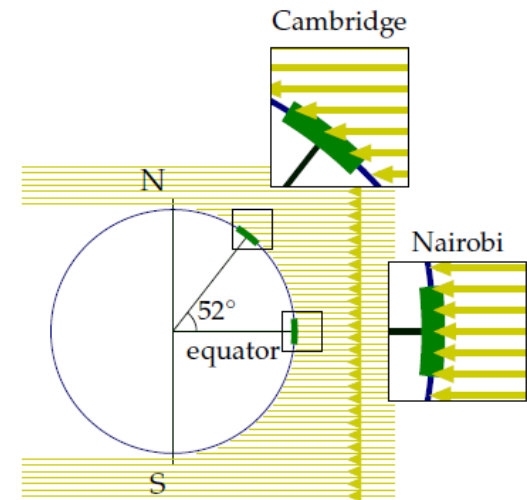
Limites do desenvolvimento/crescimento de renováveis

Energia solar

We are estimating how our consumption stacks up against conceivable sustainable production. In the last three chapters we found car-driving and plane-flying to be bigger than the plausible on-shore wind-power potential of the United Kingdom. Could solar power put production back in the lead?

The power of raw sunshine at midday on a cloudless day is 1000 W per square metre. That's 1000 W per m^2 of area oriented towards the sun, not per m^2 of land area. To get the power per m^2 of *land area* in Britain, we must make several **corrections**. We need to compensate for the tilt between the sun and the land, which reduces the intensity of midday sun to about **60%** of its value at the equator (figure 6.1). We also lose out because it is not midday all the time. On a cloud-free day in March or September, the ratio of the *average* intensity to the midday intensity is about **32%**. Finally, we lose power because of cloud cover. In a typical UK location the sun shines during just **34%** of daylight hours.

The combined effect of these three factors and the additional complication of the wobble of the seasons is that the average raw power of sunshine per square metre of south-facing roof in Britain is roughly $110 \text{ W}/\text{m}^2$, and the average raw power of sunshine per square metre of flat ground is roughly $100 \text{ W}/\text{m}^2$.



Limites do desenvolvimento/crescimento de renováveis



Energia solar: painéis fotovoltaicos

Photovoltaic (PV) panels convert sunlight into electricity. Typical solar panels have an efficiency of about 10%; expensive ones perform at 20%. (Fundamental physical laws limit the efficiency of photovoltaic systems to at best 60% with perfect concentrating mirrors or lenses, and 45% without concentration. A mass-produced device with efficiency greater than 30% would be quite remarkable.) The average power delivered by south-facing 20%-efficient photovoltaic panels in Britain would be

$$20\% \times 110 \text{ W/m}^2 = 22 \text{ W/m}^2.$$

Figure 6.5 shows data to back up this number. Let's give every person 10 m^2 of expensive (20%-efficient) solar panels and cover all south-facing roofs. These will deliver

5 kWh per day per person.

$$22 \times 10 \times 24\text{h} = 5280 \text{ W/d.pessoa}$$

Limites do desenvolvimento/crescimento de renováveis

Energia solar: painéis fotovoltaicos



Cenário:

utilizar 5% da área do país!

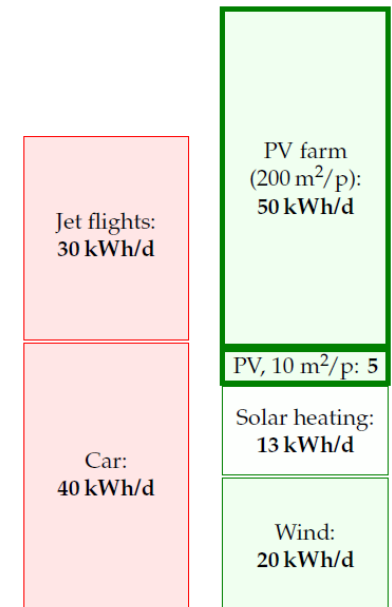
Fantasy time: solar farming

If a breakthrough of solar technology occurs and the cost of photovoltaics came down enough that we could deploy panels all over the countryside, what is the maximum conceivable production? Well, if we covered 5% of the UK with 10%-efficient panels, we'd have

$$\begin{aligned} &10\% \times 100 \text{ W/m}^2 \times 200 \text{ m}^2 \text{ per person} \\ \simeq &50 \text{ kWh/day/person.} \end{aligned}$$

I assumed only 10%-efficient panels, by the way, because I imagine that solar panels would be mass-produced on such a scale only if they were very cheap, and it's the lower-efficiency panels that will get cheap first. The power density (the power per unit area) of such a solar farm would be

$$10\% \times 100 \text{ W/m}^2 = 10 \text{ W/m}^2.$$



Energia e sustentabilidade

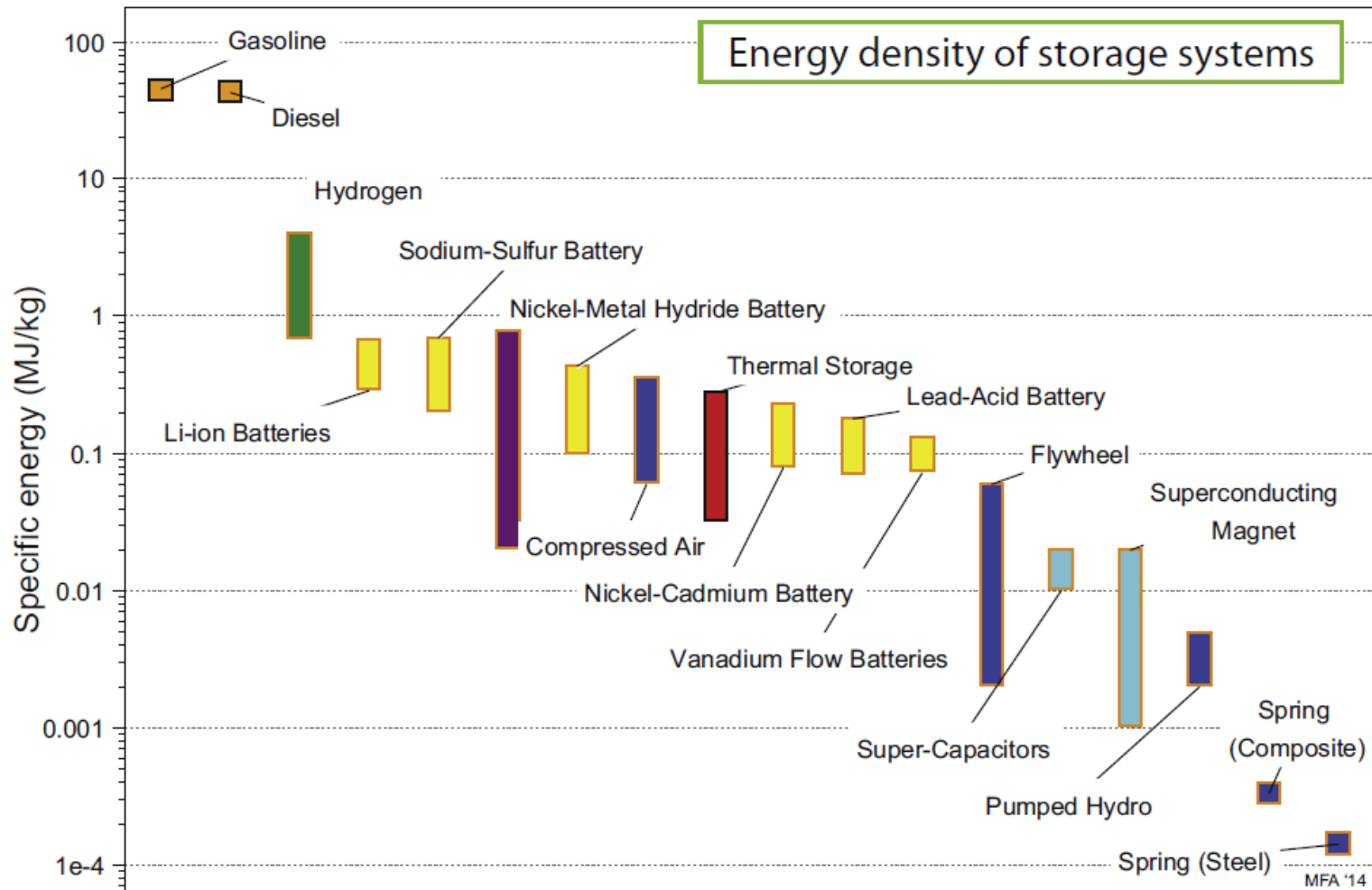
Sistemas de armazenamento de energia

- Quando a energia é gerada de uma forma intermitente e não está sincronizada com o seu consumo, esta energia tem de ser armazenada até à sua utilização.
- E este processo é um grande desafio quando lidamos com energias renováveis, que são na sua maioria intermitentes.



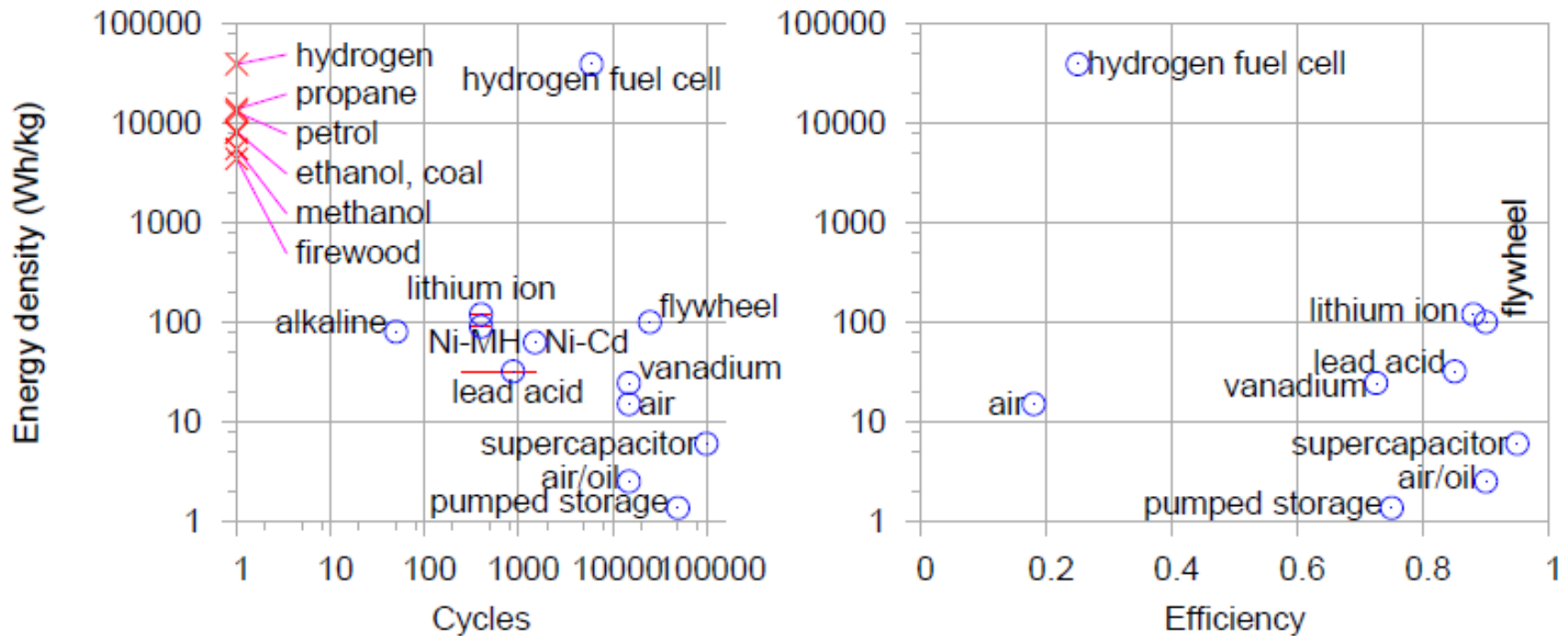
Energia e sustentabilidade

Sistemas de armazenamento de energia



Energia e sustentabilidade

Sistemas de armazenamento de energia e combustíveis



Energia e sustentabilidade

Sistemas de armazenamento de energia: comparação

Storage System	Specific Energy (MJ/kg)	Energy Density (MJ/m ³)	Operating Cost (\$/MJ)	Specific Power (W/kg)	Efficiency (%)	Cycle Life (#-cycles) ^b
Conventional fuels	20–50	15,000–72,000	-	-	18–50	1
Pumped hydro	0.002–0.005	2–5	0.0006–0.0014	0.02–0.3	70–80	400,000
Compressed air energy storage	0.36	25	0.0001–0.0019	8	65–70	15,000
Springs	0.00014–0.00033	0.6–1.1	-	-	98–99.9	<i>Depends on loading¹</i>
Flywheels	0.002–0.025	2.1	0.0008–0.0017	100–10,000	75–85	150,000
Thermal storage	0.032–0.036	160	0.0008–0.0019	1.5–1.7	72–85	10,000–15,000
Li-ion batteries	0.32–0.68	720–1,400	0.0019–0.0047	250–340	80–90	300–2,000
Sodium–sulfur batteries	0.2–0.7	140–540	0.0039	10–15	75–83	3600–4700
Lead-acid batteries	0.07–0.18	200–430	0.0008–0.0028	4–180	70–90	200–1500
Nickel cadmium batteries	0.08–0.23	72–310	0.0008–0.0055	30–150	60–85	800–1200
Nickel-metal hydride batteries	0.1–0.43	190–1300	0.0006–0.0028	4–140	65–85	300–1000
Vanadium flow batteries	0.07–0.13	110–170	0.0039	2.2–3.1	71–88	10000–16000
Hydrogen fuel cells	0.02–0.8	1–70	0.0006–0.0041	5–20	27–35	5,000–10,000
Super capacitors	0.010–0.020	10–25	0.0008–0.0019	1,000–20,000	90–95	1,000,000
Superconducting magnetic energy storage	0.001–0.02	2–10	0.0003–0.0083	500–20,000	85–92	50,000–200,000

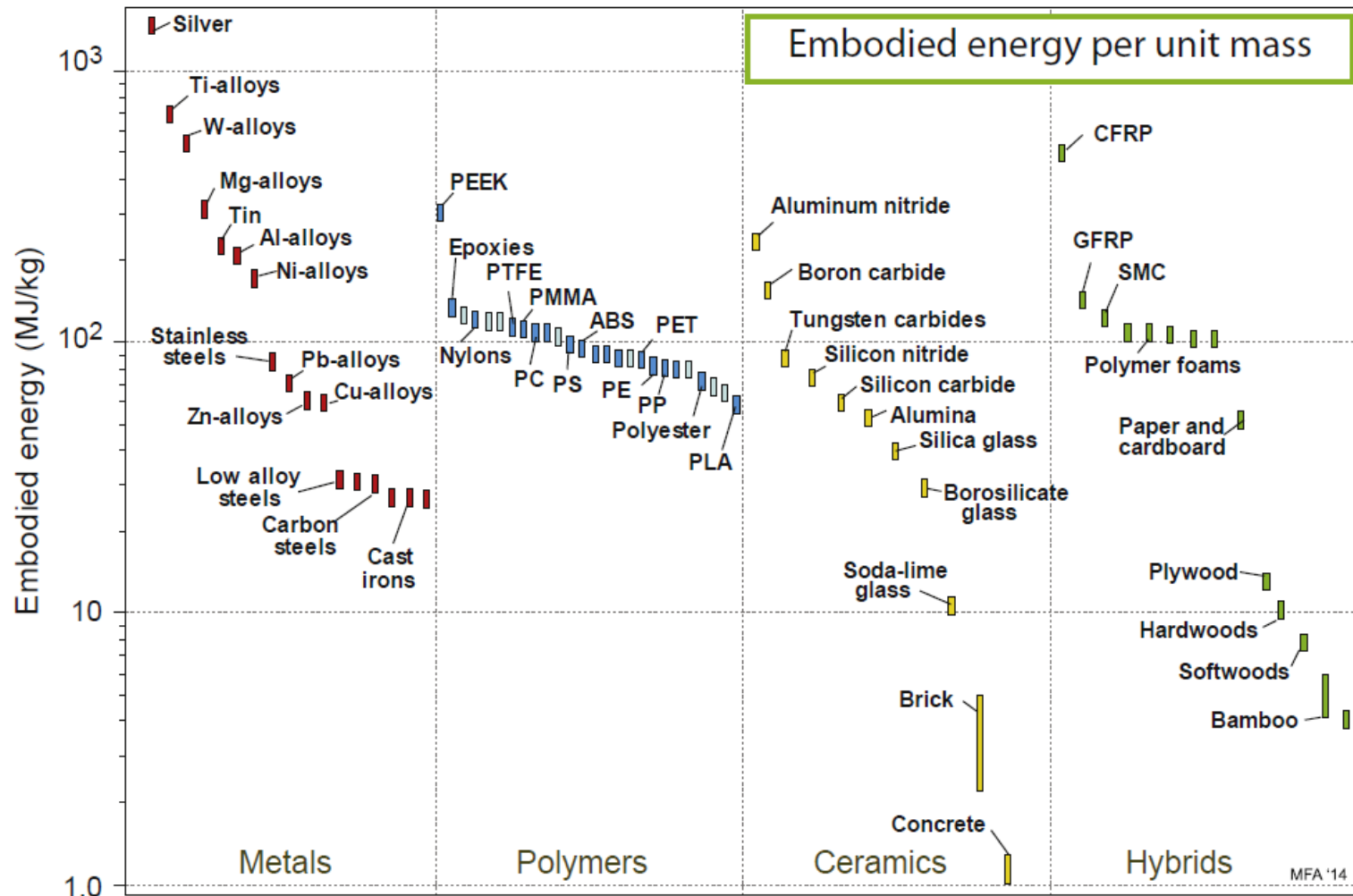
Energia e sustentabilidade

Energia incorporada na obtenção de materiais: indicador ambiental

- Todos os materiais contêm energia, usada na exploração e conformação, tratamento térmico ou químico, ou intrínseca (caso dos polímeros e elastómeros derivados do petróleo).
- Usar um material é usar a energia contida ou necessária à sua manipulação, com óbvia penalização ambiental: geração de CO_2 , óxidos nitrosos (NO_x), compostos sulfurosos, poeiras e calor perdido.
- Por isso a energia é um dos indicadores ambientais mais usados, também porque a sua contabilização é mais fácil de realizar.
- Estes índices ambientais (conteúdo energético) são derivados de forma análoga aos indicadores de desempenho (p.ex. peso mínimo) e de custo.

Energia e sustentabilidade

Energia incorporada: indicador ambiental



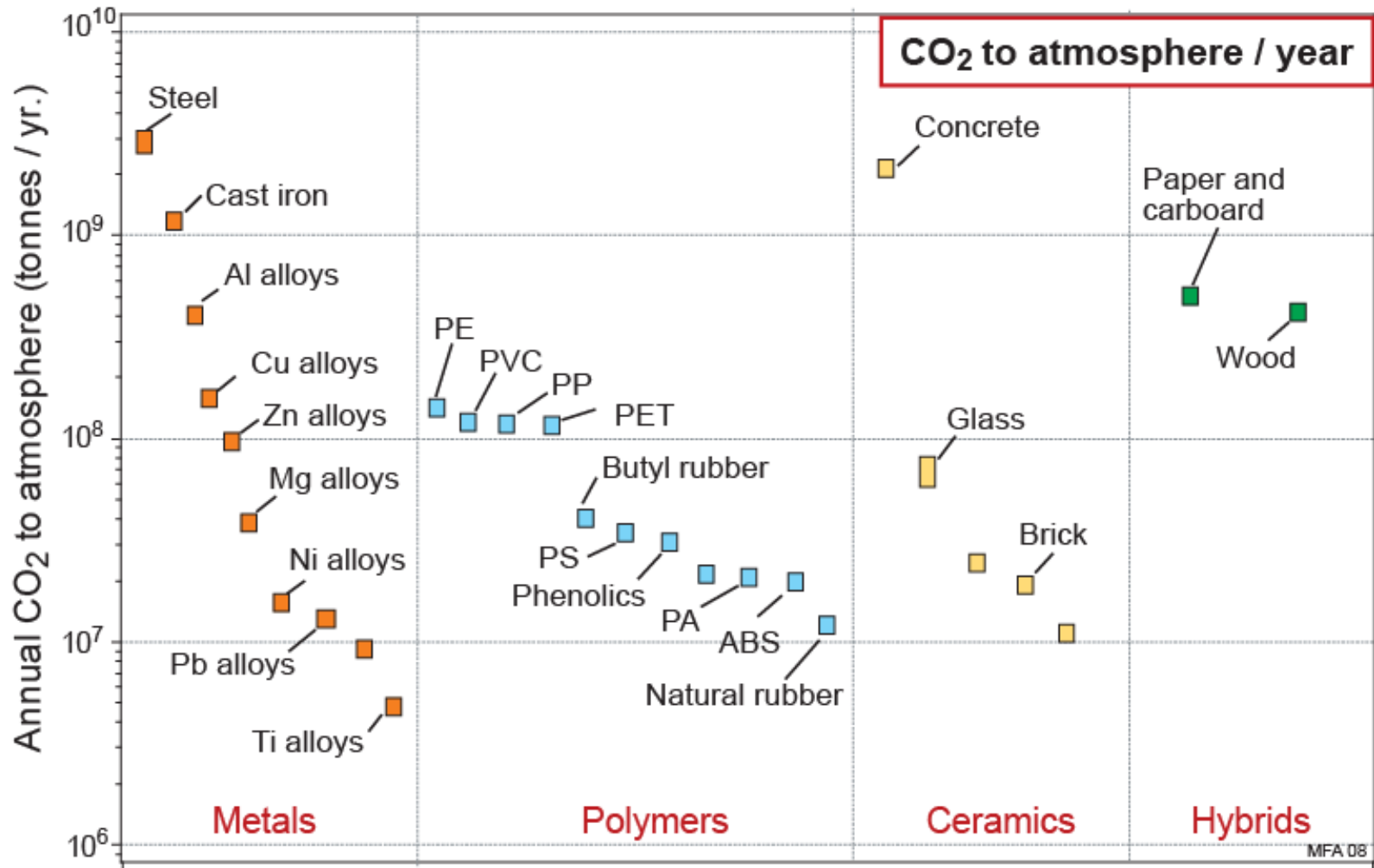
Energia incorporada: indicador ambiental

Element	Embodied Energy MJ/kg	CO _{2,eq} (GWP) kg/kg	Water Usage l/kg
Aluminum	210	13	1.2e3
Antimony	140	13	3.4e3
Beryllium	6.4e3	400	360
Bismuth	150	9.3	2.9e3
Cadmium	13	0.68	820
Calcium	150	9.5	-
Cerium	510	33	55
Chromium	300	16	470
Cobalt	130	8.3	580
Copper	60	3.7	310
Dysprosium	1.4e3	90	55
Erbium	2.9e3	190	55
Europium	20e3	1.3e3	55
Gadolinium	1.3e3	86	55
Gallium	3.1e3	210	3.8e3
Gold	250e3	15e3	-
Hafnium	1.1e3	69	8.1e3
Holmium	1.9e3	120	55
Indium	2.9e3	180	1.3e3
Iridium	45e3	2.9e3	-
Iron	45	3	50
Lanthanum	290	19	55
Lead	27	1.9	340
Lithium	590	28	470
Lutetium	48e3	3e3	55
Magnesium	320	36	980

Element	Embodied Energy MJ/kg	CO _{2,eq} (GWP) kg/kg	Water Usage l/kg
Manganese	65	4	240
Mercury	130	8.8	470
Molybdenum	380	34	360
Neodymium	85	5.5	55
Nickel	170	11	230
Niobium	1.7e3	110	150
Osmium	62e3	4e3	-
Palladium	160e3	8.5e3	-
Platinum	270e3	15e3	-
Praseodymium	650	42	55
Rhenium	12e3	770	3.8e3
Rhodium	560e3	30e3	-
Samarium	1.2e3	77	55
Scandium	59e3	3.8e3	55
Selenium	72	4.5	26
Silver	1.5e3	100	76e3
Tantalum	4.3e3	260	650
Tellurium	160	7.5	26
Terbium	9e3	580	55
Thallium	2.4e3	150	1.8e3
Thulium	26e3	1.7e3	55
Tin	230	13	11e3
Titanium	580	39	110
Tungsten	540	34	150
Uranium	1.3e3	81	3.5e3
Vanadium	3.7e3	250	650
Ytterbium	5.6e3	360	55
Yttrium	1.5e3	93	55

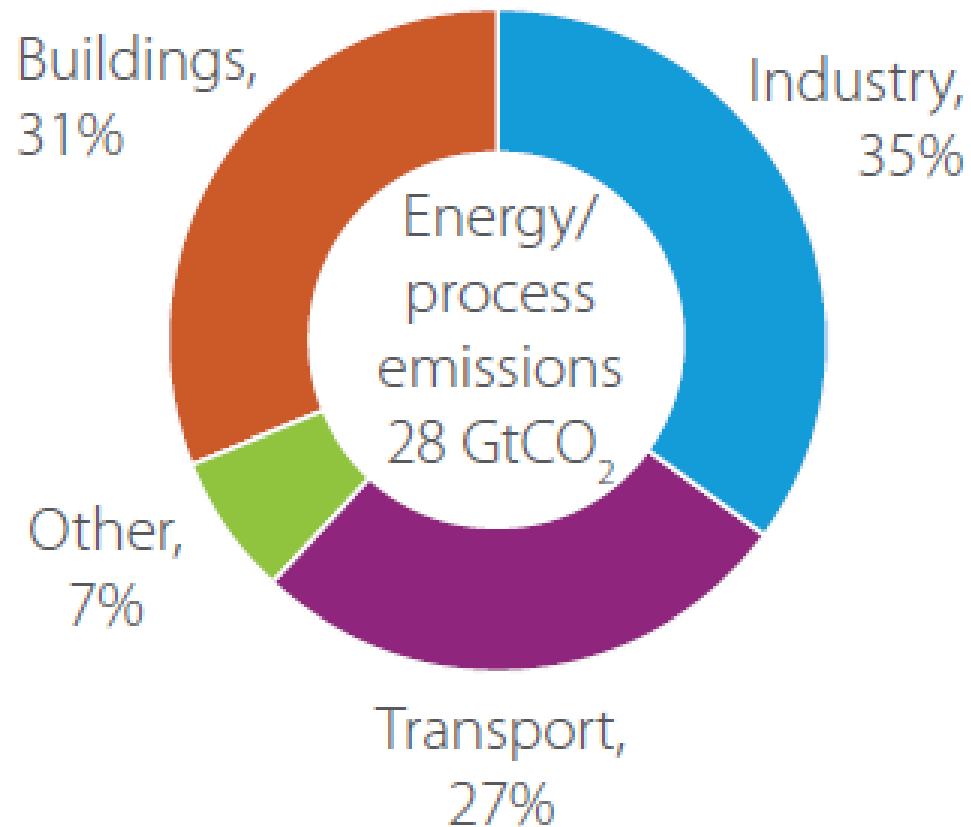
Energia e sustentabilidade

Emissões de CO₂/ano



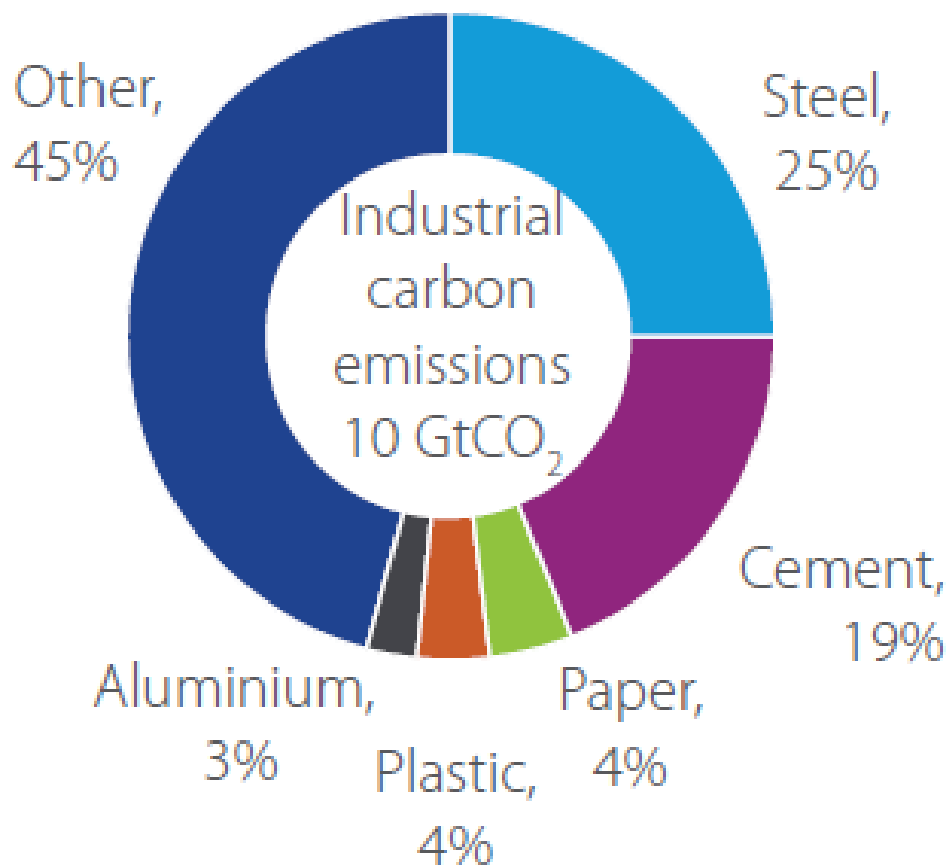
Energia e sustentabilidade

Fontes antropogénicas de CO₂ por setor



Energia e sustentabilidade

Fontes antropogénicas de CO₂ por material (com identificação de 5 materiais chave)



Energia e sustentabilidade

Can 100% renewable energy power the world?

- Federico Rosei and Renzo Rosei

